

PACKING $k^2 - c$ UNIT SQUARES: $s(k^2 - c) = k$ **FOR ALL LARGE k**

SUNGJOON RYU

ABSTRACT. Let $s(n)$ be the side of the smallest square into which n non-overlapping unit squares can be packed. We show that for every fixed $c \geq 0$ one has $s(k^2 - c) = k$ for all sufficiently large integers k . Equivalently, if $M(k)$ denotes the largest number of unit squares that can be placed in $[0, k]^2$ so that they are pairwise disjoint as closed sets, then $k^2 - M(k) \rightarrow \infty$; in fact $k^2 - M(k) \geq \kappa \log k$ for every $k \geq 2$, with an absolute constant $\kappa > 0$. The proof combines a sharpened form of the fundamental lemma of Roth and Vaughan, with an explicit constant and a further sharpening for nearly parallel pairs, with two simple observations: a closedness constraint on horizontal lines, which is special to integer sides, and a quantization of heights in columns of nearly axis-parallel squares, measured from the bottom and from the top side and organised over a continuum of scales; a count of the cracks that can break the quantization controls the columns, and an angular argument, sharpened by a computer-assisted branch-and-bound computation, bounds the overlap of the Roth–Vaughan rectangles at uncovered points. We obtain $\kappa = 0.033$. Since $k^2 - M(k) \geq 1$ for every $k \geq 2$ by an elementary argument, the logarithmic bound has content only for $\log k > 1/\kappa \approx 30$, and the result should be regarded as qualitative.

1. INTRODUCTION

For $n \geq 1$ let $s(n)$ denote the side of the smallest square that contains n unit squares with pairwise disjoint interiors (rotations allowed). Trivially $s(k^2) = k$, and $s(n) \leq \lceil \sqrt{n} \rceil$. For $n = k^2 - c$ with c small compared with k it is natural to expect that one cannot do better than the trivial $k \times k$ grid with c squares removed. Nagamochi [21] stated $s(k^2 - 1) = s(k^2 - 2) = k$ for all $k \geq 2$; the scoring assertion of his Lemma 1, on which the published proof rests, has recently been shown to fail [7, 18] (Karakuş points out that this does not disprove the rectangle bound that Nagamochi derives from it), and replacement proofs have been given: of $s(k^2 - 1) = k$ in [18], through a weaker rectangle bound proved with a strip measure, and of $s(k^2 - 2) = k$ (which implies $s(k^2 - 1) = k$), with a formalization in Lean, in [7]; see also the discussion in [10]. For $k^2 - 3$ the values $k = 3, \dots, 7$ are known [19, 3, 4] ($k = 5, 6$ in the preprint [4]), and a computer-assisted proof of $s(k^2 - 3) = k$ for all $k \geq 6$, not yet refereed, has recently been announced [10]; the same repository has since announced, again by a computer-assisted and unrefereed argument, that $s(k^2 - 4) = k$ for all $k \geq 5$ [10, directory `s12/certificates/k2m4`]. Friedman’s survey [15] contains the conjecture that $s(n^2 - c) = n$ implies $s((n+1)^2 - c) = n+1$, and Chung and Graham [8, concluding remarks] wrote that it seems likely that $s(n^2 - 1) = n$, and that the same should hold for $n^2 - c$ unit squares when c is fixed and n is sufficiently large (attributing this to [15]). On the other hand, $s(k^2 - c) < k$ as soon as c is large in terms of k : Arslanov, Mustafin and Shangitbayev [2] show $s(n^2 - n) < n$ for $n \geq 12$. Erdős and Graham [11] packed a square of side x with wasted area $O(x^{7/11})$, and this exponent was lowered several times (see the introductions of [20, 6]); the exponent $3/5$ was claimed by Chung and Graham [9]; an error in their argument was pointed out by Arslanov and Bui [1], and the bound $O(x^{3/5})$ was re-established, independently and by different methods, in the preprints of Bui [6] and of

Date: Preprint, 5 October 2026.

McClenagan [20] (the latter based on his 2024 thesis). Applied to x slightly below k , this gives $s(k^2 - Ck^{3/5}) < k$ for large k .

Following the notes [10, file `s12/search/FRIEDMAN.md`], define

$$c^*(k) := \max\{c \in \mathbb{Z}_{\geq 0} : s(k^2 - c) = k\}.$$

Since s is non-decreasing, $s(k^2 - c) = k$ holds exactly for $0 \leq c \leq c^*(k)$, and Friedman's conjecture says precisely that c^* is non-decreasing. The results above give $3 \leq c^*(k) \leq O(k^{3/5})$ for $k \geq 3$; here the lower bound uses the unrefereed announcement [10] for $k \geq 8$ (and the later announcement would give $c^*(k) \geq 4$ for $k \geq 5$), and the upper bound uses the preprints [20, 6] (the refereed bound $O(x^{(3+\sqrt{2})/7} \log x)$ of [8] gives $c^*(k) = O(k^{(3+\sqrt{2})/7} \log k)$). Whether $c^*(k) \rightarrow \infty$, i.e. whether for every fixed c one has $s(k^2 - c) = k$ for all large k , as expected in [8], is recorded as open in [10], where it is also observed that it does not follow from the lower bound of Roth and Vaughan [23]: their theorem [23, p. 170, Theorem] states that the wasted area $w(\alpha)$ in a square of side α satisfies $w(\alpha) \gg (\alpha \|\alpha\|)^{1/2}$ whenever $\alpha(\alpha - [\alpha]) > 1/6$, where $\|\alpha\|$ is the distance from α to the nearest integer, and this vanishes as $\alpha \rightarrow k^-$. We are not aware of a proof in the literature that $c^*(k)$ is unbounded.

Theorem 1.1. *There is an absolute constant $\kappa > 0$ such that for every integer $k \geq 2$, every family of N unit squares in $[0, k]^2$ that are pairwise disjoint as closed sets satisfies*

$$k^2 - N \geq \kappa \log k.$$

One may take $\kappa = 0.033$ ($\kappa = 0.027$ without the computer-assisted Lemma 4.10, see Remark 7.5; both constants of this theorem use that lemma). Moreover, for $k \geq k_2 := 10^{13}$ one has $k^2 - N \geq (0.99977 \log k + 0.3819)/30.147$. (Here and below $\log$ is the natural logarithm.)

Corollary 1.2. *$c^*(k) \geq \kappa \log k - 1$ for $k \geq 2$. In particular $c^*(k) \rightarrow \infty$: for every fixed c , $s(k^2 - c) = k$ for all sufficiently large k . This confirms the expectation of Chung and Graham [8] stated above.*

Since $k^2 - N \geq 1$ for every $k \geq 2$ (Corollary 3.3), the bound $\kappa \log k$ says something new only for $\log k > 1/\kappa \approx 30$; for $k < k_2$ we have $0.033 \log k < 0.99$, so there the theorem is just Corollary 3.3, and the threshold k_2 only serves to make the constants in the proof simple (see Remark 7.3).

Idea of the proof. On a horizontal line the squares cut out pairwise disjoint closed intervals of $[0, k]$, so at most $k - 1$ of them have length ≥ 1 (Lemma 3.2). A square of inclination a is cut in an interval of length $\sec a \geq 1$ except near its top and bottom (“ramps”, of height $\sin a$). Hence, unless the line carries some waste, it must either cross a square of large inclination, which by a sharpened form of the Roth–Vaughan fundamental lemma (Proposition 4.3) costs waste proportional to the inclination, or cross a nearly axis-parallel square through its ramp. In the latter case we look down the column below that square: if the column is “rigid” (little vertical waste, only tiny inclinations) and avoids the narrow “ends” of the squares in it, heights in it are quantized, i.e. the square's corners sit within $1.08 \cdot 10^{-5}$ of integers, so its ramp cannot meet a line at a height whose fractional part is farther than that from 0 and 1. A rigid column through an end of a square must pass through a tight crack between two squares, and at most one crack per unit of height fits under a square; so the exceptional columns have measure about $1/C$, where $1/(Cs)$ is the inclination threshold for columns at scale s . Thus many columns must be non-rigid, which again costs waste. Since k is an integer, the same holds for columns hanging from the top side. A line at height y is treated with the inclination threshold $1/(C'd(y))$, where $d(y)$ is the distance to the nearer of the bottom and the top side, and each column is charged only at the least scale at which it fails to be rigid; integrating over heights y with $\sqrt{k}/4 \leq d(y) \lesssim k/2$ with the measure $dy/d(y)$ then gives a lower bound of order $\log k$; within this scheme the

choice $C' = 2\sqrt{2}$, $C = 2$ is optimal. Two further refinements matter for the constant: on a line with little waste the short chords cut from ramps add up to almost a full unit, for nearly parallel, nearly touching squares the fundamental lemma holds with a much better constant (Proposition 4.5), and only the uncovered points of the Roth–Vaughan rectangles matter, where at most 13 of them can overlap (Lemma 4.9), and at most 9 by a computer-assisted refinement (Lemma 4.10). Only the numerical constants ($\kappa = 0.033$ and the bound for $k \geq k_2$) depend on the computer: with Lemma 4.9 alone the proof gives $\kappa = 0.027$ (Remark 7.5).

2. CLOSED PACKINGS AND THE REDUCTION

Throughout, a *square* is a closed unit square. A *closed packing* of $[0, k]^2$ is a finite family of squares contained in $[0, k]^2$ that are pairwise disjoint as closed sets. Put $M(k)$ for the maximum size of a closed packing of $[0, k]^2$ and, for a given closed packing, $W := k^2 - N$, the area of $[0, k]^2$ not covered by the squares.

Lemma 2.1. *For integers $k \geq 2$ and $n \geq 1$: $s(n) < k$ if and only if $n \leq M(k)$. Moreover $M(k) \leq k^2 - 1$ (Corollary 3.3 below), so $c^*(k) = k^2 - M(k) - 1$.*

Proof. The minimum in the definition of $s(n)$ is attained by compactness. If n squares with disjoint interiors lie in a square of side $s < k$, place that square concentrically in $[0, k]^2$, let O be the common centre, and replace each centre c_i by $O + \lambda(c_i - O)$ with $\lambda = k/s > 1$, keeping orientations. If two of the original squares are separated by a line with normal u , the separation $u \cdot (c_2 - c_1) \geq h_1(u) + h_2(u)$ (where $h_i(u)$ is the half-width of the i -th square in direction u) becomes strict after dilation, and every moved square stays in $[0, k]^2$: in coordinates centred at O , $|c_i \cdot e| + h_i(e) \leq s/2$ for e a coordinate direction implies $\lambda|c_i \cdot e| + h_i(e) \leq \lambda s/2 - (\lambda - 1)h_i(e) \leq k/2$. So we obtain a closed packing of $[0, k]^2$ with n squares. Conversely, a closed packing has a positive minimum distance η between its squares; contracting all centres about the centre of $[0, k]^2$ by a factor $\mu < 1$ close to 1 keeps the squares disjoint and places them in a concentric square of side $\mu k + (1 - \mu)\sqrt{2} < k$. Finally, $s(n) \leq k$ always holds for $n \leq k^2$, so for integers $0 \leq c < k^2$ we have $s(k^2 - c) = k$ iff $k^2 - c > M(k)$. $\square$

Thus Theorem 1.1 implies Corollary 1.2. Note that W is an integer.

3. ELEMENTARY GEOMETRY

The *inclination* $a(S) \in [0, \pi/4]$ of a square S is the least angle between a side of S and a side of $[0, k]^2$; the inclination $\theta(S_1, S_2)$ between two squares is the least angle between their sides (this is the quotient metric on $\mathbb{R}/(\pi/2)\mathbb{Z}$, with values in $[0, \pi/4]$), and we write $S_0 := \partial[0, k]^2$ with $\theta(S, S_0) := a(S)$. These are the definitions of [23]. Likewise, for a single side σ of $[0, k]^2$ we put $\theta(S, \sigma) = \theta(\sigma, S) := a(S)$.

Lemma 3.1. *Let S be a square with inclination a and lowest point at height v .*

- (a) *The vertical extent of S is $[v, v + \cos a + \sin a]$.*
- (b) *For $y \in \mathcal{M}(S) := [v + \sin a, v + \cos a]$ (the middle band, of length $\cos a - \sin a$) the line $y = \text{const}$ meets S in a segment of length $\sec a \geq 1$; for y in the ramp set $\mathcal{R}(S) := [v, v + \sin a] \cup (v + \cos a, v + \cos a + \sin a]$ the length is $< \sec a$. If $a = 0$, then $\mathcal{R}(S) = \emptyset$.*
- (c) *Symmetrically, for x in the vertical middle band $\mathcal{M}'(S)$, an interval of length $\cos a - \sin a$ (defined in the proof; for $a = 0$ it is the horizontal extent of S), the vertical line through x meets S in a segment of length $\sec a$, entering S through a side of slope $\pm \tan a$; the lower endpoint of that segment has height in $[v, v + \sin a]$. The complement of $\mathcal{M}'(S)$ in the horizontal extent of S (the vertical ramp set) has total length $2 \sin a$.*

Proof. For $a = 0$ everything is clear. Let $0 < a \leq \pi/4$ and (up to reflection) let the vertices be $P_0 = (x_0, v)$, $P_1 = P_0 + (\cos a, \sin a)$, $P_3 = P_0 + (-\sin a, \cos a)$, $P_2 = P_1 + P_3 - P_0$, at heights v , $v + \sin a$, $v + \cos a$, $v + \cos a + \sin a$. For $y \in [v + \sin a, v + \cos a]$ the left and right boundary points lie on the parallel sides P_0P_3 and P_1P_2 , which are at distance 1 and have direction $(-\sin a, \cos a)$; hence the horizontal chord is $1/\cos a$. Outside this band one end of the chord lies on P_0P_1 or P_3P_2 , and the chord is shorter. For (c), put $\mathcal{M}'(S) := [x_0, x_0 + \cos a - \sin a]$ (in the reflected position, its mirror image $[x_0 - \cos a + \sin a, x_0]$). On this interval the lower and upper boundary points lie on P_0P_1 and P_3P_2 , parallel sides at distance 1 with direction $(\cos a, \sin a)$, so the vertical chord is $\sec a$; the lower point has height $v + (x - x_0)\tan a \in [v, v + (\cos a - \sin a)\tan a] \subset [v, v + \sin a]$. The horizontal extent has length $\cos a + \sin a$, so the vertical ramp set has length $2\sin a$. $\square$

For $y \in [0, k]$ let $\omega(y)$ be the length of the part of the segment $[0, k] \times \{y\}$ not covered by squares (the *line waste*); by Fubini $\int_0^k \omega(y) dy = W$. Similarly, for a vertical segment σ we call the length of σ not covered by squares its *vertical waste*; integrating the vertical waste of $\{x\} \times [0, k]$ over x also gives W . Since a packing is finite, all functions of y (or of x) considered below are piecewise continuous with finitely many pieces, and all sets of y (or of x) that we define are finite unions of intervals; we shall not comment on measurability again.

Lemma 3.2 (closedness). *Let $k \geq 2$. On every horizontal line, the squares of a closed packing of $[0, k]^2$ meet the segment $[0, k] \times \{y\}$ in pairwise disjoint closed intervals, and at most $k - 1$ of these have length ≥ 1 .*

Proof. Disjointness is inherited from the squares. If k of the intervals had length ≥ 1 , their union would be a closed subset of $[0, k]$ of full measure, hence all of $[0, k]$, which is then a disjoint union of $k \geq 2$ nonempty closed sets, contradicting connectedness. $\square$

Corollary 3.3. *For $k \geq 2$ every closed packing of $[0, k]^2$ has $W \geq 1$; that is, $M(k) \leq k^2 - 1$.*

Proof. W is an integer. If $W = 0$, then $\omega(y) = 0$ for some (indeed almost every) y , so the intervals cut out on that line cover $[0, k]$ up to a null set; each has length $\leq \sqrt{2} < k$, so there are at least two, and as in the proof of Lemma 3.2 their union is a closed set of full measure, hence $[0, k]$, a contradiction. $\square$

For a square S and $y \in \mathbb{R}$ let $c_S(y)$ be the length of the horizontal chord $S \cap (\mathbb{R} \times \{y\})$ (zero if the line misses S).

Lemma 3.4 (short chords). *Let $k \geq 2$, $y \in (0, k)$ and $0 < \alpha \leq \pi/4$, and suppose that every square meeting the line y has inclination $< \alpha$. Then*

$$\sum_{S: 0 < c_S(y) < 1} c_S(y) > 1 - \omega(y) - (k - 1)(\sec \alpha - 1),$$

the sum being over the squares S of the packing. Moreover, for a square S let $\Phi(S) := \{y : 0 < c_S(y) < 1\}$. If $a := a(S) > 0$, then $\Phi(S) \subset \mathcal{R}(S)$, $|\Phi(S)| = 2\sin a \cos a$ and $\int_{\Phi(S)} c_S(y) dy = \sin a \cos a$; if $a(S) = 0$, then $\Phi(S) = \emptyset$.

Proof. By Lemma 3.1(b), a chord of length ≥ 1 has length at most $\sec a(S) < \sec \alpha$, and by Lemma 3.2 there are at most $k - 1$ such chords on the line; the chords cover a length $k - \omega(y)$ of $[0, k]$, which gives the inequality. For the second part let $a > 0$ and let v be the height of the lowest point of S ; in the coordinates of the proof of Lemma 3.1, for $v \leq y \leq v + \sin a$ the chord joins the sides P_0P_3 and P_0P_1 and has length $(y - v)(\tan a + \cot a) = (y - v)/(\sin a \cos a)$, which is < 1 exactly for $y < v + \sin a \cos a$; symmetrically at the top, and in the middle band the chord is $\sec a \geq 1$. Hence $\Phi(S)$ consists of two intervals of length $\sin a \cos a$ in $\mathcal{R}(S)$, on each of which

c_S grows linearly from 0 to 1, so its integral there is $\frac{1}{2} \sin a \cos a$. If $a = 0$, every nonzero chord has length 1. $\square$

4. A SHARPENED FUNDAMENTAL LEMMA AND THE COST OF INCLINATIONS

Throughout this section $k \geq 4$ (this is used only in Lemma 4.8(b), which is not needed for the main theorem), a closed packing of $[0, k]^2$ is fixed, $S_0 := \partial[0, k]^2$, and the *waste* is the set of points of $(0, k)^2$ that lie in no square; its area is W . We begin with two elementary facts about gaps between convex sets.

Lemma 4.1 (convex gaps). *Let I be an interval of length λ , let $0 \leq \varphi \leq 1$, and let $g \geq 0$ be a continuous, convex, piecewise linear function on I with finitely many break points, such that $|g'| \geq \varphi$ off the break points and the slope of g increases by at least 2 at every break point. Then $\int_I g \geq \varphi(2 - \varphi)\lambda^2/4$.*

Proof. We may assume $\varphi > 0$ and $\lambda > 0$. Let y^* be a minimum point of g on the closure of I , and let a and $b = \lambda - a$ be the lengths of the parts of I to the left and to the right of y^* . If y^* is an end point of I , then g is monotone with $|g'| \geq \varphi$, so $g(y) \geq \varphi|y - y^*|$ and $\int_I g \geq \varphi\lambda^2/2 \geq \varphi(2 - \varphi)\lambda^2/4$. Otherwise y^* is a break point, since $g' \neq 0$ elsewhere; let $-p \leq 0 \leq q$ be the slopes of g just to the left and just to the right of y^* . Then $p, q \geq \varphi$ and $p + q \geq 2$, and by convexity $g(y) \geq p(y^* - y)$ for $y < y^*$ and $g(y) \geq q(y - y^*)$ for $y > y^*$. By the Cauchy–Schwarz inequality, $\lambda^2 = (a + b)^2 \leq (pa^2 + qb^2)(1/p + 1/q)$, so

$$\int_I g \geq \frac{pa^2 + qb^2}{2} \geq \frac{\lambda^2}{2(1/p + 1/q)}.$$

Let, say, $p \leq q$. If $q \geq 2 - \varphi$, then $1/p + 1/q \leq 1/\varphi + 1/(2 - \varphi)$. If $q < 2 - \varphi$, then $q \geq 1$ (as $p + q \geq 2$) and $p \geq 2 - q$, so $1/p + 1/q \leq F(q) := 1/(2 - q) + 1/q$; the function F is convex on $(0, 2)$ and symmetric about 1, so $F(q) \leq F(2 - \varphi) = 1/\varphi + 1/(2 - \varphi)$. In both cases $1/p + 1/q \leq 2/(\varphi(2 - \varphi))$, and the lemma follows. $\square$

Let (s, t) be Cartesian coordinates in the plane. For a square Z let $[\tau^-(Z), \tau^+(Z)]$ be its projection to the t -axis, and for $\tau^-(Z) \leq t \leq \tau^+(Z)$ let $l_Z(t)$ and $r_Z(t)$ be the least and the largest s with $(s, t) \in Z$. The vertices of Z at which $t = \tau^+(Z)$ or $t = \tau^-(Z)$ are its *t-extreme* vertices.

Lemma 4.2 (boundary functions). *The function l_Z is convex and r_Z is concave. Each of them is piecewise linear with at most one break point, at which the slope of l_Z increases, and that of r_Z decreases, by at least 2. Off the break points $l'_Z(t) = \tan \psi$, where $\psi \in (-\frac{\pi}{2}, \frac{\pi}{2})$ is the angle between the t -axis and the side of Z containing $(l_Z(t), t)$, and similarly for r_Z . Moreover $\tau^+(Z) - \tau^-(Z) \geq 1$.*

Proof. If a side of Z is parallel to the t -axis, then l_Z and r_Z are constant and $\tau^+(Z) - \tau^-(Z) = 1$. Otherwise let w be the vertex of Z with the least s . The two sides at w have directions $(\cos \gamma, \sin \gamma)$ and $(\sin \gamma, -\cos \gamma)$ for some $0 < \gamma < \frac{\pi}{2}$. The point $(l_Z(t), t)$ lies on the first side for $t \geq t(w)$ and on the second for $t \leq t(w)$, where ds/dt equals $\cot \gamma = \tan(\frac{\pi}{2} - \gamma)$ and $-\tan \gamma = \tan(-\gamma)$, respectively; the slope increases at $t(w)$ by $\cot \gamma + \tan \gamma = 1/(\sin \gamma \cos \gamma) \geq 2$, and l_Z is convex. The case of r_Z is symmetric, and $\tau^+(Z) - \tau^-(Z) = \cos \gamma + \sin \gamma \geq 1$. $\square$

For a square S_1 and S_2 either another square or S_0 , let L be the segment from the centre c_1 of S_1 to the centre c_2 of S_2 , respectively to a point c_2 of S_0 nearest to c_1 (any choice will do), and let $R = R(S_1, S_2)$ be the closed rectangle of points at distance at most $\frac{1}{2}$ from the line through L whose orthogonal projection lies on L . This is the rectangle used by Roth and Vaughan [23, §3, p. 172]; the following proposition replaces their constant $c = 10^{-10}$ by an explicit, much

larger one. Let u be the unit vector along L and n a unit normal, and use the coordinates (s, t) of $c_1 + su + tn$; thus $R = \{0 \leq s \leq |L|, |t| \leq \frac{1}{2}\}$.

Proposition 4.3 (sharpened fundamental lemma). *Let $\theta := \theta(S_1, S_2)$, and let m' be the number of squares $Z \notin \{S_1, S_2\}$ of the packing that have a t -extreme vertex $v \in R$ with $|t(v)| < \frac{1}{2}$. Let $\mathcal{T} \subset (-\frac{1}{2}, \frac{1}{2})$ be the set of the numbers $t(v)$ for these vertices, and let $\lambda_1, \dots, \lambda_q$ be the lengths of the intervals into which the points of $\mathcal{T}$ cut $(-\frac{1}{2}, \frac{1}{2})$. Then the waste in R has area at least $\frac{\theta(2-\theta)}{4} \sum_r \lambda_r^2 \geq \theta(2-\theta)/(4(m'+1))$.*

Proof. Centres of squares lie in the convex set $[\frac{1}{2}, k - \frac{1}{2}]^2$. If S_2 is a square, L lies in this set and R lies within $\frac{1}{2}$ of it, so $R \subset [0, k]^2$. If $S_2 = S_0$, then L is perpendicular to a side of $[0, k]^2$ nearest to c_1 , and R is the rectangle of width 1 standing on that side, centred at the foot of L ; since $c_1 \in [\frac{1}{2}, k - \frac{1}{2}]^2$, again $R \subset [0, k]^2$, and $[0, k]^2$ lies in the half-plane $\{s \leq |L|\}$. Let $\Sigma := \{|t| < \frac{1}{2}\}$ be the open strip, λ_t the line at height t , and $\ell_t := \lambda_t \cap R$ for $|t| < \frac{1}{2}$. The points $c_1 + tn$ with $|t| < \frac{1}{2}$ are at distance less than $\frac{1}{2}$ from c_1 , hence interior points of S_1 ; likewise the points $c_2 + tn$ are interior points of S_2 if S_2 is a square.

(a) *Squares in the strip.* Let $Z \notin \{S_1, S_2\}$ be a square that meets some chord ℓ_{t_0} , $|t_0| < \frac{1}{2}$. The convex set $Z \cap \Sigma$ contains no interior point of S_1 or S_2 , so it misses the segment $\Sigma \cap \{s = 0\}$ and, if S_2 is a square, the segment $\Sigma \cap \{s = |L|\}$; it therefore lies on one side of each of them, and as it meets ℓ_{t_0} , we get $Z \cap \Sigma \subset R$ (in the wall case we also use $Z \subset \{s \leq |L|\}$). Hence for $|t| < \frac{1}{2}$ the square Z meets ℓ_t if and only if $\tau^-(Z) \leq t \leq \tau^+(Z)$, in the coordinates (s, t) . If $|\tau^+(Z)| < \frac{1}{2}$, a t -extreme vertex of Z at which $t = \tau^+(Z)$ lies in $Z \cap \Sigma \subset R$, so Z is one of the m' squares of the statement; the same holds for $\tau^-(Z)$. Since $\tau^+(Z) - \tau^-(Z) \geq 1$ (Lemma 4.2), at most one of $\tau^\pm(Z)$ lies in $(-\frac{1}{2}, \frac{1}{2})$.

Consequently the numbers $\tau^\pm(Z)$ in $(-\frac{1}{2}, \frac{1}{2})$, for all such Z , are exactly the points of $\mathcal{T}$, and $\mathcal{T}$ has at most m' elements. Cutting $(-\frac{1}{2}, \frac{1}{2})$ at the points of $\mathcal{T}$ gives $q \leq m' + 1$ open intervals I_r of lengths λ_r , with $\sum_r \lambda_r = 1$.

(b) *The same squares in the same order.* Fix r . By (a), for every square $Z \notin \{S_1, S_2\}$ either Z meets ℓ_t for all $t \in I_r$ or for none. Two disjoint compact convex sets that both meet all the lines λ_t , $t \in I_r$, are strictly separated by a line that is not parallel to u , so they appear in the same order on all these lines. Hence for $t \in I_r$ the chord ℓ_t meets the same squares in the same order, say $Z_0 = S_1, Z_1, \dots, Z_M$, followed by $Z_{M+1} = S_2$ (or the wall S_0). For $1 \leq i \leq M+1$ and $t \in I_r$ let

$$g_i(t) := l_{Z_i}(t) - r_{Z_{i-1}}(t) \geq 0$$

(with $l_{Z_{M+1}} \equiv |L|$ in the wall case); this is the length of the uncovered part of ℓ_t between Z_{i-1} and Z_i , since by (a) the chord of each Z_i , $1 \leq i \leq M$, on the line λ_t lies in ℓ_t . The regions swept by these uncovered parts for different i and r are disjoint parts of the waste in R .

(c) *Each gap is convex.* By Lemma 4.2, g_i is convex and piecewise linear on I_r , and at each of its break points its slope increases by at least 2. Off the break points, $g'_i = \tan \psi' - \tan \psi$, where $\psi, \psi' \in (-\frac{\pi}{2}, \frac{\pi}{2})$ are the angles between n and the sides of Z_{i-1} and Z_i (or the wall) bounding the gap, so

$$|g'_i| \geq |\psi' - \psi| \geq \varphi_i := \theta(Z_{i-1}, Z_i),$$

because the direction of a side of a square is congruent modulo $\pi/2$ to the orientation of the square. As $\varphi_i \leq \frac{\pi}{4} < 1$, Lemma 4.1 gives $\int_{I_r} g_i \geq \varphi_i(2 - \varphi_i)\lambda_r^2/4$.

(d) *Summation.* By the triangle inequality for inclinations, $\sum_{i=1}^{M+1} \varphi_i \geq \theta$. Put $\varphi'_i := \min(\varphi_i, \theta)$; then $\sum_i \varphi'_i \geq \theta$ (either some $\varphi_i \geq \theta$, or $\varphi'_i = \varphi_i$ for all i), and since $x(2-x)$ is

increasing on $[0, 1]$,

$$\sum_i \varphi_i(2 - \varphi_i) \geq \sum_i \varphi'_i(2 - \varphi'_i) \geq (2 - \theta) \sum_i \varphi'_i \geq \theta(2 - \theta).$$

Summing over i and r , the waste in R has area at least

$$\sum_r \frac{\theta(2 - \theta)\lambda_r^2}{4} \geq \frac{\theta(2 - \theta)}{4q} \left(\sum_r \lambda_r \right)^2 \geq \frac{\theta(2 - \theta)}{4(m' + 1)}. \quad \square$$

For nearly parallel, nearly touching pairs the bound $\sum_r \lambda_r^2 \geq 1/(m' + 1)$ is far from the truth: the cut points can only come from squares wedged into the two small triangles left free by S_1 and S_2 in R , and few of them fit there. We first record an elementary fact about sections of a square.

Lemma 4.4 (sections near a vertex). *Let Z be a square, e a unit vector, and v a vertex of Z at which $p \mapsto p \cdot e$ is smallest on Z . For $0 \leq \nu \leq \frac{3}{4}$ the line $\{p : p \cdot e = v \cdot e + \nu\}$ meets Z in a segment of length at least $\min(2\nu, 1)$.*

Proof. Measure heights by $p \cdot e - v \cdot e$, and let $0 \leq \psi \leq \frac{\pi}{2}$ be the angle between one side at v and the line $\{p \cdot e = v \cdot e\}$; the neighbours of v are at heights $\sin \psi$ and $\cos \psi$, and the opposite vertex at $\sin \psi + \cos \psi \geq 1$. If $\psi \in \{0, \frac{\pi}{2}\}$ every section at height $\nu \in (0, 1]$ has length 1. Otherwise the section at height ν has length $\nu/(\sin \psi \cos \psi) \geq 2\nu$ for $\nu \leq \min(\sin \psi, \cos \psi)$, length $1/\max(\sin \psi, \cos \psi) \geq 1$ for ν between $\min(\sin \psi, \cos \psi)$ and $\max(\sin \psi, \cos \psi)$, and length $(\sin \psi + \cos \psi - \nu)/(\sin \psi \cos \psi)$ for larger ν . In the last case, if $\nu \leq \frac{3}{4}$, then $\sin \psi, \cos \psi \leq \frac{3}{4}$; with $p := \sin \psi + \cos \psi \in [1, \sqrt{2}]$ we get $p - \sin \psi \cos \psi = 1 - \frac{1}{2}(p - 1)^2 \geq 0.91 > \nu$, so the length exceeds 1. $\square$

Proposition 4.5 (nearly parallel pairs). *Let S_1 be a square and S_2 a square or S_0 , at distance less than d , with $\theta := \theta(S_1, S_2) \leq \theta_1$, where $d, \theta_1 \leq 10^{-5}$. Then $\sum_r \lambda_r^2 \geq Q_* := 0.32$ in Proposition 4.3, so the waste in $R(S_1, S_2)$ has area at least $Q_*\theta(2 - \theta)/4$.*

Proof. Put $\varepsilon_1 := d + \theta_1 \leq 2 \cdot 10^{-5}$ and $\varepsilon_0 := 10^{-4}$. We use the notation of the proof of Proposition 4.3; recall that every square Z contributing a point of $\mathcal{T}$ satisfies $Z \cap \Sigma \subset R$ and $Z \cap (S_1 \cup S_2) = \emptyset$. A point $\tau \in \mathcal{T}$ is *deep* if $|\tau| \leq \frac{1}{2} - \varepsilon_0$. For each deep τ fix a square Z with $\tau^-(Z) = \tau$ or $\tau^+(Z) = \tau$; we call τ an *upper* point in the first case, when Z lies in $\{t \geq \tau\}$, with *depth* $\frac{1}{2} - \tau$, and a *lower* point in the second case, with *depth* $\tau + \frac{1}{2}$. Different points come from different squares, since at most one of $\tau^\pm(Z)$ lies in $(-\frac{1}{2}, \frac{1}{2})$. For t in the t -range of a set X let $J_X(t)$ be the section of X by the line $\lambda_t = \{p : t(p) = t\}$.

(a) *Shallow points.* Let λ'_r be the lengths of the intervals into which the deep points cut $(-\frac{1}{2}, \frac{1}{2})$. Adding the shallow points (those in $(\frac{1}{2} - \varepsilon_0, \frac{1}{2})$ or $(-\frac{1}{2}, -\frac{1}{2} + \varepsilon_0)$) cuts only the top and the bottom interval, say of lengths $m_+ \leq 1$ and $m_- \leq 1$; parts of length at least $m_+ - \varepsilon_0$ and $m_- - \varepsilon_0$ remain uncut, so $\sum_r \lambda_r^2 \geq \sum_r \lambda_r'^2 - 4\varepsilon_0$ (if top and bottom interval coincide, the bound is $(1 - 2\varepsilon_0)^2$).

(b) *The wall case.* Let $S_2 = S_0$. For each side of $[0, k]^2$ the distance from S_1 to that side equals the distance from c_1 to it minus $\frac{1}{2}(\cos \theta + \sin \theta)$, so the side on which R stands is also a side nearest to S_1 . In coordinates in which $R = [-\frac{1}{2}, \frac{1}{2}] \times [0, |L|]$ stands on the x -axis and $t(p) = x$, S_1 has inclination θ and centre $(0, |L|)$, so it contains the square $(0, |L|) + [-\frac{\sigma_\theta}{2}, \frac{\sigma_\theta}{2}]^2$ with $\sigma_\theta := 1/(\cos \theta + \sin \theta) \geq 1 - \theta$; its lowest point is at height $< d$, so $|L| < d + \frac{1}{2}(\cos \theta + \sin \theta)$ and $R \setminus S_1 \subset \{y < |L| - \frac{\sigma_\theta}{2}\} \cup \{|x| > \frac{\sigma_\theta}{2}\}$, where $|L| - \frac{\sigma_\theta}{2} < d + \theta \leq \varepsilon_1$. If $\tau = \tau^-(Z)$ were deep, Lemma 4.4 would give, for $0 < \nu \leq \varepsilon_0/2$, a section of Z of length $\geq 2\nu$ on the line $x = \tau + \nu$, which satisfies $|x| \leq \frac{1}{2} - \frac{\varepsilon_0}{2} < \frac{1 - \theta}{2} \leq \frac{\sigma_\theta}{2}$ (as $\theta < \varepsilon_0$); this section lies in $R \setminus S_1$, hence in $\{y < \varepsilon_1\}$,

which is impossible for $\nu = \varepsilon_0/2 > \varepsilon_1/2$. The same holds for τ^+ . So there are no deep points, and $\sum \lambda_r^2 \geq (1 - 2\varepsilon_0)^2 > Q_*$.

(c) *Normal form.* Let now S_2 be a square. Two disjoint convex polygons are separated by a line parallel to a side of one of them (the origin lies outside the convex polygon $S_1 - S_2$, whose sides are parallel to sides of S_1 or S_2 , so it violates one of the inequalities defining that polygon). Exchanging S_1 and S_2 if necessary (this does not change R or $\sum \lambda_r^2$) and choosing coordinates, we may assume $S_1 = [-\frac{1}{2}, \frac{1}{2}]^2$, $S_2 \subset \{x \geq \frac{1}{2}\}$, and $c_2 = (a, b)$ with $b \geq 0$. Then S_2 has inclination θ , its projections to the axes have length $w := \cos \theta + \sin \theta \in [1, 1 + \theta]$, and S_2 contains the axis-parallel square $S_2^\circ := c_2 + [-\frac{\sigma_\theta}{2}, \frac{\sigma_\theta}{2}]^2$, $\sigma_\theta := 1/w \geq 1 - \theta$. Since $S_2 \subset \{x \geq \frac{1}{2}\}$ and some point of S_2 is within d of S_1 , we have $1 \leq a < 1 + d + \frac{\theta}{2}$, $0 \leq b < 1 + d + \frac{\theta}{2}$, and $\ell := a - \frac{\sigma_\theta}{2}$, the abscissa of the left side of S_2° , satisfies $\frac{1}{2} \leq \ell < \frac{1}{2} + d + \theta$. Put $h := b/a \in [0, 1 + \varepsilon_1]$ and $\rho_h := \sqrt{1 + h^2}$, so that $|L| = a\rho_h$, and take $n = (-b, a)/|L|$, so that $t(x, y) = (ay - bx)/|L|$. A direct computation gives the levels of the vertices $V_1 = (\frac{1}{2}, \frac{1}{2})$, $V_2 = (-\frac{1}{2}, \frac{1}{2})$, $V_3 = (\frac{1}{2}, -\frac{1}{2})$ of S_1 and $V'_1 = (\ell, b - \frac{\sigma_\theta}{2})$, $V'_2 = (\ell, b + \frac{\sigma_\theta}{2})$, $V'_3 = (\ell + \sigma_\theta, b - \frac{\sigma_\theta}{2})$ of S_2° :

$$\begin{aligned} t(V_1) &= \frac{1-h}{2\rho_h}, & t(V_2) &= \frac{1+h}{2\rho_h}, & t(V_3) &= -\frac{1+h}{2\rho_h}, \\ t(V'_1) &= -\sigma_\theta \frac{1-h}{2\rho_h}, & t(V'_2) &= \sigma_\theta \frac{1+h}{2\rho_h}, & t(V'_3) &= -\sigma_\theta \frac{1+h}{2\rho_h}. \end{aligned}$$

On the line λ_t , with s as coordinate, the lines $x = \frac{1}{2}$, $x = \ell$, $y = \frac{1}{2}$, $y = b - \frac{\sigma_\theta}{2}$ are met at $s = \frac{\rho_h}{2} + ht$, $s = \rho_h \ell + ht$, $s = (\frac{\rho_h}{2} - t)/h$ and $s = (\rho_h(b - \frac{\sigma_\theta}{2}) - t)/h$ respectively ($h > 0$).

(d) *The free gap.* For $|t| < \frac{1}{2}$ the points $c_1 + tn$ and $c_2 + tn$ lie in the interiors of S_1 and S_2 , and $J_R(t)$ is the segment between them. Hence, if Z contributes a point of $\mathcal{T}$ and $J_Z(t) \neq \emptyset$ with $|t| < \frac{1}{2}$, then $J_Z(t) \subset J_R(t) \setminus (S_1 \cup S_2)$ lies between $J_{S_1}(t)$ and $J_{S_2}(t) \supset J_{S_2^\circ}(t)$, and $|J_Z(t)| \leq \Gamma(t)$, where $\Gamma(t)$ denotes the distance from the right end of $J_{S_1}(t)$ to the left end of $J_{S_2^\circ}(t)$. The right end of $J_{S_1}(t)$ lies on $x = \frac{1}{2}$ for $t(V_3) \leq t \leq t(V_1)$ and on $y = \frac{1}{2}$ for $t(V_1) \leq t \leq t(V_2)$; the left end of $J_{S_2^\circ}(t)$ lies on $y = b - \frac{\sigma_\theta}{2}$ for $t(V'_3) \leq t \leq t(V'_1)$ and on $x = \ell$ for $t(V'_1) \leq t \leq t(V'_2)$. If $h \leq 1$, put $t_{\text{lo}} := t(V'_1)$ and $t_{\text{hi}} := \min(t(V_1), t(V'_2))$; for $t \in [t_{\text{lo}}, t_{\text{hi}}]$ the two ends lie on $x = \frac{1}{2}$ and $x = \ell$, so $\Gamma(t) = \rho_h(\ell - \frac{1}{2})$. If $h > 1$, put $t_{\text{lo}} := t(V_1) < 0$ and $t_{\text{hi}} := t(V'_1)$; for $t \in [t_{\text{lo}}, t_{\text{hi}}]$ they lie on $y = \frac{1}{2}$ and $y = b - \frac{\sigma_\theta}{2}$, so $\Gamma(t) = \rho_h(b - \frac{\sigma_\theta}{2} - \frac{1}{2})/h$. In both cases $\Gamma \leq \gamma_* := 1.415\varepsilon_1$ on $[t_{\text{lo}}, t_{\text{hi}}]$, since $\ell - \frac{1}{2} < \varepsilon_1$, $b - \frac{\sigma_\theta}{2} - \frac{1}{2} < \varepsilon_1$, $\rho_h \leq 1.415$, and $\rho_h/h \leq \sqrt{2}$ for $h > 1$.

(e) *Depths.* Let $\tau = \tau^-(Z)$ be an upper point and suppose $\tau < t_{\text{hi}} - 1.5\gamma_*$. Put $t' := \tau + 0.75\gamma_*$ if this is $\geq t_{\text{lo}}$, and $t' := t_{\text{lo}}$ otherwise. Then $t' \in [t_{\text{lo}}, t_{\text{hi}}]$, $|t'| < \frac{1}{2}$, and $0.75\gamma_* \leq t' - \tau < \frac{1}{2}$ (note $t_{\text{lo}} \leq 0$ and $\tau \geq -\frac{1}{2} + \varepsilon_0$), so by Lemma 4.4 and (d), $\gamma_* \geq \Gamma(t') \geq |J_Z(t')| \geq \min(2(t' - \tau), 1) \geq 1.5\gamma_*$, a contradiction. Hence every upper point has depth at most $D_u := \frac{1}{2} - t_{\text{hi}} + 1.5\gamma_*$, and symmetrically (using $t' := \tau - 0.75\gamma_*$ or t_{hi}) every lower point has depth at most $D_l := \frac{1}{2} + t_{\text{lo}} + 1.5\gamma_*$; in particular $D_u + D_l = 1 - (t_{\text{hi}} - t_{\text{lo}}) + 3\gamma_* \leq 1 + \beta_1$ with $\beta_1 := 3\gamma_*$. Let $D(h) := \frac{1}{2} - \frac{1-h}{2\rho_h}$; it is increasing, with derivative $(1+h)/(2\rho_h^3)$, and $D(h) \geq \frac{1}{2}$ for $h \geq 1$. By (c), if $h \leq 1$ then $\frac{1}{2} - t(V_1) = D(h)$, $\frac{1}{2} - t(V'_2) \leq \frac{1}{2} - \frac{1+h}{2\rho_h} + \frac{1+h}{2\rho_h}\theta \leq 0.71\theta$ (as $(1+h)/\rho_h \leq \sqrt{2}$ for $0 \leq h \leq 1$) and $\frac{1}{2} + t(V'_1) \leq D(h) + \frac{\theta}{2}$, while if $h > 1$ then $\frac{1}{2} - t(V'_1)$ and $\frac{1}{2} + t(V_1)$ are at most $\frac{1}{2} \leq D(h)$; so in all cases $D_u, D_l \leq D(h) + 0.71\theta + 1.5\gamma_* \leq D(h) + 3\varepsilon_1$ (as $\theta \leq \varepsilon_1$ and $1.5\gamma_* < 2.13\varepsilon_1$).

(f) *Two points on the same side.* Suppose $D(h) \geq 0.19$; then $h \geq 0.344$ (as $D(0.344) < 0.19$), so that $\tau_* := t(V'_2) - \frac{1}{2} > 0$ and $t(V_1) \leq t(V'_2) \leq t(V_2)$. Let Z give an upper point τ of depth ζ . Then

$$t(V'_2) - \tau = \zeta + \tau_* \leq D_u + \tau_* \leq D(h) + \frac{1+h}{2\rho_h} - \frac{1}{2} + 3\varepsilon_1 = \frac{h}{\rho_h} + 3\varepsilon_1 < \frac{3}{4},$$

and $\tau \geq \frac{1}{2} - D_u > -0.02$. Hence for $t \in [\tau, t(V'_2)]$ each of Z , S_1 , S_2° has a nonempty section on λ_t (the t -ranges of S_1 and S_2° contain $[t(V'_3), t(V'_2)] \supset [-0.6, t(V'_2)]$, and that of Z contains $[\tau, \tau + 1]$); the end points of these sections depend continuously on t and the three sets are pairwise disjoint, so their order on λ_t does not depend on t . By (d), applied at a level in $[\tau, \frac{1}{2}]$, $J_Z(t)$ lies between $J_{S_1}(t)$ and $J_{S_2^\circ}(t)$ for all $t \in [\tau, t(V'_2)]$. At the level $t(V'_2)$ the section of S_2° is the point V'_2 , and that of S_1 ends on the side $y = \frac{1}{2}$; by (c) the gap between them is

$$\Gamma_* := \rho_h \left(\ell - \frac{1}{2} \right) + \frac{\rho_h}{2h} (h - 1 + \sigma_\theta(1 + h)) \leq \rho_h + \rho_h \left(\ell - \frac{1}{2} \right) < 2,$$

and by Lemma 4.4 the section $J_Z(t(V'_2))$ has length at least $\min(2(\zeta + \tau_*), 1)$. Now let two upper points of depths $\zeta' \leq \zeta$ come from squares Z', Z . Their sections at the level $t(V'_2)$ are disjoint and lie in a segment of length Γ_* ; hence $\zeta' + \tau_* < \frac{1}{2}$. If $\zeta + \tau_* \leq \frac{1}{2}$, then $2(\zeta' + \tau_*) + 2(\zeta + \tau_*) \leq \Gamma_*$, that is, $\zeta' \leq \chi_u - \zeta$ with $\chi_u := \frac{1}{2}\Gamma_* - 2\tau_*$. If $\zeta + \tau_* > \frac{1}{2}$, then $2(\zeta' + \tau_*) + 1 \leq \Gamma_*$, so

$$\zeta' \leq \frac{\Gamma_*}{2} - t(V'_2) = \frac{h(h-1)}{2\rho_h} + \frac{\rho_h}{2} \left(\ell - \frac{1}{2} \right) - \frac{(1-\sigma_\theta)(1+h)(1-h)^2}{4h\rho_h} < 1.1\varepsilon_1 < \varepsilon_0,$$

which is impossible for a deep point; so in this case there is at most one upper point. With $\chi(h) := 1 + \frac{\rho_h}{2} - \frac{1+h}{\rho_h}$ we get $\chi_u \leq \chi(h) + (1-\sigma_\theta)\frac{1+h}{\rho_h} + \frac{\rho_h}{2}(\ell - \frac{1}{2}) \leq \chi(h) + 2.2\varepsilon_1$. The lower points are treated in the same way below the strip, down to the level $-\frac{1}{2} - \tau_* = t(V'_3)$, where the section of S_2° is the point V'_3 and that of S_1 ends on $x = \frac{1}{2}$ (as $t(V_3) \leq t(V'_3) \leq t(V_1)$), at distance $\Gamma_l := \rho_h(a - \frac{1-\sigma_\theta}{2}) \leq \rho_h + \rho_h(a-1)$. For two lower points of depths $\zeta' \leq \zeta$ this gives $\zeta' \leq \chi_l - \zeta$ if $\zeta + \tau_* \leq \frac{1}{2}$, where $\chi_l := \frac{1}{2}\Gamma_l - 2\tau_* \leq \chi(h) + (1-\sigma_\theta)\frac{1+h}{\rho_h} + \frac{\rho_h}{2}(a-1) \leq \chi(h) + 2.2\varepsilon_1$, and $\zeta' \leq \frac{1}{2}\Gamma_l - \frac{1}{2} - \tau_* \leq \frac{h(h-1)}{2\rho_h} + \frac{\rho_h}{2}(a-1) + (1-\sigma_\theta)\frac{1+h}{2\rho_h} < 1.8\varepsilon_1 < \varepsilon_0$ otherwise.

(g) *Conclusion.* Let $\zeta_u \geq \zeta'_u$ ($\zeta_l \geq \zeta'_l$) be the largest and the second largest depth of an upper (lower) point, with the value 0 if there is no such point. Consider the intervals $(\frac{1}{2} - \zeta_u + \beta_1, \frac{1}{2} - \zeta'_u)$, $(-\frac{1}{2} + \zeta'_l, -\frac{1}{2} + \zeta_l - \beta_1)$ and $(-\frac{1}{2} + \zeta_l, \frac{1}{2} - \zeta_u)$ (some may be empty). They contain no deep point: upper points have depth ζ_u , ζ'_u or less than ζ'_u , and lower points lie at levels $\leq -\frac{1}{2} + D_l \leq \frac{1}{2} - D_u + \beta_1 \leq \frac{1}{2} - \zeta_u + \beta_1$; symmetrically for the second interval. They are pairwise disjoint, by the same inequality. Hence, with $\mu_u := \zeta_u - \zeta'_u$ and $\mu_l := \zeta_l - \zeta'_l$,

$$\sum_r \lambda_r^2 \geq (1 - \zeta_u - \zeta_l)_+^2 + (\mu_u - \beta_1)_+^2 + (\mu_l - \beta_1)_+^2 \geq (1 - \zeta_u - \zeta_l)_+^2 + \mu_u^2 + \mu_l^2 - 2.04\beta_1,$$

since $(\mu - \beta_1)_+^2 \geq \mu^2 - 2\beta_1\mu$ and $\mu \leq D_u, D_l \leq 0.51$. If $D(h) < 0.19$, then $\zeta_u, \zeta_l < 0.1906$ and $\sum_r \lambda_r^2 \geq (1 - 0.3812)^2 > 0.38$, so by (a) $\sum_r \lambda_r^2 \geq 0.38 - 4\varepsilon_0 > Q_*$. Let $D(h) \geq 0.19$ and $\chi_* := \max(\chi_u, \chi_l)$. By (f), on each side either $\mu = \zeta$ (at most one point) or $\mu \geq (2\zeta - \chi_*)_+$; in both cases $\mu^2 \geq \xi(\zeta) := \min\{\zeta^2, ((2\zeta - \chi_*)_+)^2\}$. Hence $\sum_r \lambda_r^2 \geq \Xi(\zeta_u, \zeta_l) - 2.04\beta_1$ with $\Xi(z, z') := (1 - z - z')_+^2 + \xi(z) + \xi(z')$. The function χ decreases on $[0, h_c]$ and increases on $[h_c, \infty)$, where $h_c^3 + 3h_c - 2 = 0$ (indeed $\chi'(h) = (h^3 + 3h - 2)/(2\rho_h^3)$), so $\chi_* \leq \max(\chi(0.344), \chi(1.00002)) + 2.2\varepsilon_1 < 0.29295$. If $z + z' > 1$, then $z, z' > 0.49 > \chi_*$ (as $z, z' \leq \max(D_u, D_l) \leq D(h) + 3\varepsilon_1 \leq 0.51$ by (e)) and $\Xi(z, z') \geq z^2 + z'^2 > 0.48$. Otherwise, choosing one of the two expressions for each of $\xi(z)$ and $\xi(z')$, Ξ is bounded below by the least of: $(1 - z - z')^2 + z^2 + z'^2 \geq \frac{1}{3}$; $(1 - z - z')^2 + z^2 + ((2z' - \chi_*)_+)^2 \geq \frac{1}{2}(1 - z')^2 + ((2z' - \chi_*)_+)^2 \geq \frac{1}{9}(2 - \chi_*)^2$ (the minimum is at $z' = \frac{1+4\chi_*}{9}$; for $z' \leq \frac{\chi_*}{2}$ the bound is $\frac{1}{2}(1 - \frac{\chi_*}{2})^2$, which is larger); and $(1 - z - z')^2 + ((2z - \chi_*)_+)^2 + ((2z' - \chi_*)_+)^2 \geq \frac{2}{3}(1 - \chi_*)^2$ (a convex symmetric function, minimal at $z = z' = \frac{1+2\chi_*}{6}$). For $\chi_* < 0.29295$ all these exceed 0.3237, so $\sum_r \lambda_r^2 \geq 0.3237 - 2.04\beta_1 > 0.3235$ (as $\beta_1 = 3\gamma_* \leq 8.5 \cdot 10^{-5}$). By (a), $\sum_r \lambda_r^2 \geq 0.3235 - 4\varepsilon_0 > Q_*$. $\square$

Remark 4.6. Roth and Vaughan [23, §3, pp. 172–173] assume that S_1 and S_2 are at distance at most 1, bound the number m of squares meeting R by 20, and use a single strip of width $1/(4m+1)$ parallel to L that contains no vertex in its interior, and a single trapezoid in it; this gives the lower bound $(4m+1)^{-2} \tan(\theta/2m)$, about $3.8 \cdot 10^{-6} \theta$ for $m = 20$, and their lemma is stated with $c = 10^{-10}$ [23, p. 171, (3)]. In Proposition 4.3 the strip is cut only where the set of squares met by the chords changes, the gaps are allowed to be convex rather than affine, and all sub-bands and gaps are summed; with the count of Lemma 4.8(a) this gives $\theta(2-\theta)/40 \geq \theta/33$ for pairs at distance less than 10^{-4} , and Proposition 4.5 gives $0.32 \theta(2-\theta)/4 \geq \theta/6.26$ for pairs at distance less than 10^{-5} that are moreover nearly parallel. In randomized adversarial searches (with S_2 perturbed as in Proposition 4.5) we found no configuration with $\sum \lambda_r^2 < \frac{1}{3}$; the value $\frac{1}{3}$ is attained by two squares wedged into the two triangles at depth $\frac{1}{3}$, and two deep points never seemed to share a triangle. Roth and Vaughan’s standing assumption $\alpha(\alpha - [\alpha]) > 1/6$ on the side of the container plays no role here. A version of their lemma for squares joined by a path, with a constant derived from theirs, is given in [5, Section 4].

Let $\mathcal{V}_h$ consist of the squares of the packing together with the left and the right side of $[0, k]^2$. Two distinct members P, Q of $\mathcal{V}_h$ are *consecutive on the line y* ($0 < y < k$) if both meet the segment $[0, k] \times \{y\}$ and the open segment between them on that line meets no square. Let G_h be the graph on $\mathcal{V}_h$ in which P and Q are adjacent if they are consecutive on a set of lines y of positive measure. Define $\mathcal{V}_v$ and G_v in the same way, with vertical lines and the bottom and the top side of $[0, k]^2$.

Lemma 4.7 (visibility graphs are planar). *The graphs G_h and G_v are planar.*

Proof. We treat G_h . For every edge $e = \{P, Q\}$ choose a line $y_e \in (0, k)$ on which P and Q are consecutive, with $y_e \neq y_{e'}$ for $e \neq e'$ (each edge has a set of such lines of positive measure). Let P be to the left of Q on y_e , and let $a_e \in P$ and $b_e \in Q$ be the end points of the segment between them; $a_e \neq b_e$ if both are squares, since they are disjoint (if one of them is a side, possibly $a_e = b_e$, and the middle piece of the path below is empty). Represent a square by its centre, the left side by $o_- := (-2, k/2)$ and the right side by $o_+ := (k+2, k/2)$, and draw e as the path from the representative of P to a_e , along $[a_e, b_e]$, and on to the representative of Q . Here the piece inside a square is the segment from its centre to the boundary point; it lies in the square, and all its points except the last one are interior points. The piece for the left side is the segment from o_- to $(-1, y_e)$ followed by the segment from $(-1, y_e)$ to $(0, y_e) = a_e$, and symmetrically for the right side. Two such paths meet only in common end points. Indeed, segments from the centre of a square to distinct boundary points (the points a_e, b_e of different edges lie on different lines) meet only at the centre, and different squares are disjoint. The open segments (a_e, b_e) lie on distinct lines and meet no square. The pieces for the sides lie outside $(0, k) \times \mathbb{R}$, the horizontal ones lie on distinct lines, and the segments from o_- to the distinct points $(-1, y_e)$ meet only at o_- (likewise at o_+). So this is a plane drawing of G_h . $\square$

For $0 < d \leq 10^{-4}$ put $\Lambda = \Lambda(d) := d + \sqrt{2}$ and $R_0 = R_0(d) := \sqrt{\Lambda^2 + \frac{1}{4}} < 1.5001$. We use the following theorem of Fodor [13, Theorem 1]: *the smallest circle that contains 12 points with mutual distances at least 1 has radius $R_{12} = 1.5148\dots$* (explicitly, $R_{12} = 1/(\sqrt{3}x_0)$, where x_0 is the smallest positive root of $9x^5 - 15x^4 + 7x^3 - 3x + 1$). We also use Oler’s inequality [22, Theorem 1] in the following Euclidean form, which is inequality (1) of Folkman and Graham [14], where it also receives a short proof: *a compact convex set $X \subset \mathbb{R}^2$ contains at most $\frac{2}{\sqrt{3}}A(X) + \frac{1}{2}P(X) + 1$ points with mutual distances at least 1*, where A and P denote area and perimeter. (Oler’s theorem is stated for the Minkowski distance of a convex, centrally symmetric Jordan curve, with the area divided by the critical determinant, which is $\sqrt{3}/2$ for the Euclidean

norm, and for finite sets that lie in the closed region bounded by a Jordan polygon all of whose vertices belong to the set. The Euclidean form above follows by applying it to the boundary of the convex hull of the points, whose area and perimeter do not exceed those of X ; collinear points are trivial. Folkman and Graham [14, p. 751] make the same convex-hull reduction from their own inequality for planar simplicial complexes, which gives an independent proof of the form above.) Put

$$K := 27, \quad q_0 := 10.$$

Lemma 4.8 (counting). *Let $0 < d \leq 10^{-4}$.*

- (a) *If S_1 is a square and S_2 is a square or S_0 , at distance $< d$, then $m' \leq q_0 - 1 = 9$ in Proposition 4.3; hence the waste in $R(S_1, S_2)$ has area at least $\theta(2 - \theta)/(4q_0)$, $\theta = \theta(S_1, S_2)$.*
- (b) *Let $k \geq 4$ and let G be G_h or G_v . For an edge $e = \{P, Q\}$ of G whose members are at distance $< d$ put $R_e := R(P, Q)$ if both are squares, and $R_e := R(P, S_0)$ if P is a square and Q a side. Then every point lies in R_e for at most K such edges e (the two sides of $[0, k]^2$ that are vertices of G are at distance $k > d$).*

Proof. Every point of a unit square is within $\frac{\sqrt{2}}{2}$ of its centre, so the centres of two squares at distance $< d$ are at distance $< \Lambda$, and the centre of a square at distance $< d$ from S_0 is at distance $< d + \frac{\sqrt{2}}{2}$ from S_0 .

(a) For each of the m' squares Z of Proposition 4.3 choose a vertex v_Z as there, and let $Q_Z := Z \cap B(v_Z, 1)$; in coordinates in which $Z = [0, 1]^2$ and $v_Z = 0$ this is the quarter-disc $\{x, y \geq 0, x^2 + y^2 \leq 1\}$, of area $\frac{\pi}{4}$. The sets Q_Z , S_1 and (if it is a square) S_2 are pairwise disjoint, and they lie in the closed 1-neighbourhood $R + B(1)$ of R , because $v_Z, c_1, c_2 \in R$ and $S_i \subset B(c_i, \frac{\sqrt{2}}{2})$. By Steiner's formula, $R + B(1)$ has area $|L| + 2(|L| + 1) + \pi$. If S_2 is a square, then $|L| < \Lambda$ and $\frac{\pi}{4}m' + 2 \leq 3|L| + 2 + \pi$, so $m' \leq \frac{12}{\pi}|L| + 4 < \frac{12}{\pi}\Lambda + 4 < 10$. If $S_2 = S_0$, then $|L| < d + \frac{\sqrt{2}}{2}$, and all these sets lie in $[0, k]^2$, hence in the closed half-plane $\mathcal{H}$ that is bounded by the line through the side of $[0, k]^2$ on which R stands and contains $[0, k]^2$. The part of $R + B(1)$ in $\mathcal{H}$ has area $3|L| + 1 + \frac{\pi}{2}$ (we remove a 1×1 rectangle and two quarter-discs of radius 1 beyond that line), whence $\frac{\pi}{4}m' + 1 \leq 3|L| + 1 + \frac{\pi}{2}$ and $m' \leq \frac{12}{\pi}|L| + 2 < 5$. Now apply Proposition 4.3.

(b) If $p \in R_e$, the projection of p on the line of L lies on L , so p is within $\sqrt{|L|^2 + \frac{1}{4}}$ of both end points of L . For an edge joining two squares, both centres therefore lie in the open disc $B(p, R_0)$. For an edge joining a square P and a side Q , the square is at distance $< d$ from $S_0 \supset Q$, so (a) applies to (P, S_0) , and the centre of P lies within $\sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}} < R_0$ of p . Disjoint squares contain the disjoint open discs of radius $\frac{1}{2}$ about their centres, so the centres are at mutual distance at least 1; as $R_0 < R_{12}$, at most 11 squares have their centre in $B(p, R_0)$. Recall that a simple planar graph with v vertices has at most $\max(1, 3v - 6)$ edges, and that the edges in question are edges of the planar graph G (Lemma 4.7). If none of them contains a side of $[0, k]^2$, they span at most 11 vertices, so there are at most $3 \cdot 11 - 6 = 27 = K$ of them.

Otherwise let $e_0 = \{P_0, Q_0\}$ be one of them with Q_0 a side, and let σ_0 be the side of $[0, k]^2$ on which $R_{e_0} = R(P_0, S_0)$ stands. Then p is at distance at most $|L_{e_0}| < d + \frac{\sqrt{2}}{2}$ from the line through σ_0 , while every centre is at distance at least $\frac{1}{2}$ from that line, on the same side as $[0, k]^2$. Hence, if ν is the unit normal of σ_0 pointing away from $[0, k]^2$, every centre z in question satisfies $(z - p) \cdot \nu < c_0 := d + \frac{\sqrt{2}}{2} - \frac{1}{2} < 0.20721$, so all these centres lie in the compact convex set $D_0 := \{z \in \overline{B}(p, R_0) : (z - p) \cdot \nu \leq c_0\}$. With $\phi := \arccos(c_0/R_0)$, this set has area $R_0^2(\pi - \phi) + c_0\sqrt{R_0^2 - c_0^2} < 4.155$ and perimeter $2R_0(\pi - \phi) + 2\sqrt{R_0^2 - c_0^2} < 8.1$, so by Oler's inequality it contains at most $\frac{2}{\sqrt{3}} \cdot 4.155 + \frac{8.1}{2} + 1 < 9.85$, that is, at most 9, centres. Moreover,

only one side of $[0, k]^2$ occurs among the edges in question: if $\{P, Q\}$ is one of them with Q the left side, then the centre of P is within $d + \frac{\sqrt{2}}{2}$ of the left side and p is within $\sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}}$ of that centre, so p is within 1.58 of the left side; likewise for the right side (or for the bottom and the top side if $G = G_v$), and $k \geq 4 > 3.16$. So the edges span at most $9 + 1$ vertices, and there are at most $3 \cdot 10 - 6 = 24 < K$ of them. $\square$

Only the uncovered points of the rectangles R_e will matter below, and there the multiplicity is much smaller.

Lemma 4.9 (multiplicity at uncovered points). *Let $0 < d \leq 10^{-4}$, let G be G_h or G_v , and define R_e for the edges e of G whose members are at distance $< d$ as in Lemma 4.8(b). Then every point of the waste lies in R_e for at most $K_w := 13$ such edges.*

Proof. Let p be a point of the waste, $\Lambda := d + \sqrt{2}$, and call an edge e of G *relevant* if its members are at distance $< d$ and $p \in R_e$; let L_e be the segment defining R_e . Every square contains the closed disc of radius $\frac{1}{2}$ about its centre, so every centre is at distance $> \frac{1}{2}$ from p . As in the proof of Lemma 4.8(b), for a relevant e the orthogonal projection of p on the line of L_e lies on L_e , and p is at distance $\leq \frac{1}{2}$ from that line. If both members of e are squares, then $|L_e| < \Lambda$ and p is within $\sqrt{(\Lambda/2)^2 + \frac{1}{4}}$ of the nearer end point of L_e ; if e joins a square P and a side, then $|L_e| < d + \frac{\sqrt{2}}{2}$ and p is within $\sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}}$ of the centre of P . As $\Lambda/2 \leq d + \frac{\sqrt{2}}{2}$, in both cases some member of e is a square whose centre lies within $\rho_1 := \sqrt{(d + \frac{\sqrt{2}}{2})^2 + \frac{1}{4}} < 0.8662$ of p ; we fix one such member and call its centre the *inner centre* of e . Put $f(r_1, r_2) := (r_1^2 + r_2^2 - 1)/(2r_1r_2)$; if two points are at distances r_1, r_2 from a point z and at distance ≥ 1 from each other, the cosine of the angle they subtend at z is at most $f(r_1, r_2)$.

(a) *Degree at a centre.* Let c be the centre of a square P , $r := |p - c|$, and let $e_1, \dots, e_m$ be the relevant edges containing P . Let c_j be the end point of L_{e_j} other than c : the centre of the other member, or, if the other member is a side, the point of S_0 used to define $R_{e_j} = R(P, S_0)$. The angle ψ_j between $p - c$ and $c_j - c$ satisfies $\cos \psi_j \geq 0$ and $r \sin \psi_j \leq \frac{1}{2}$, so $\psi_j \leq \arcsin \frac{1}{2r}$, and all directions $c_j - c$ lie in an arc of length $2 \arcsin \frac{1}{2r}$. At most one e_j joins P to a side: the two sides of $[0, k]^2$ that are vertices of G are at distance k , while P has width at most $\sqrt{2} < k - 2d$ in the perpendicular direction. If c_i and c_j are centres, they are at distances in $[1, \Lambda)$ from c (squares at distance $< d$) and at distance ≥ 1 from each other. For fixed $r_2 \geq 1$ the function $r_1 \mapsto f(r_1, r_2) = \frac{r_1}{2r_2} + \frac{r_2^2 - 1}{2r_2} \cdot \frac{1}{r_1}$ is convex, and symmetrically in r_2 , so the maximum of f on $[1, \Lambda]^2$ is attained at a corner, where f takes the values $\frac{1}{2}$, $\frac{\Lambda}{2}$ and $1 - \frac{1}{2\Lambda^2}$; hence the directions of $c_i - c$ and $c_j - c$ differ by at least $\gamma_0 := \arccos(1 - \frac{1}{2\Lambda^2})$. If c_i lies on a side σ of $[0, k]^2$, then $c_i - c$ is a positive multiple of the unit normal ν of σ pointing away from $[0, k]^2$, c is at distance less than $d + \frac{\sqrt{2}}{2}$ from the line of σ , and every centre c_j is at distance at least $\frac{1}{2}$ from it, on the same side; so $(c_j - c) \cdot \nu < d + \frac{\sqrt{2}}{2} - \frac{1}{2}$, and as $|c_j - c| \geq 1$ the two directions differ by more than $\arccos(d + \frac{\sqrt{2}}{2} - \frac{1}{2}) > 78^\circ > \gamma_0$. Consequently

$$m \leq D(r) := 1 + \left\lfloor \frac{2 \arcsin(1/(2r))}{\gamma_0} \right\rfloor,$$

and $D(r) \leq 5$ for $r > \frac{1}{2}$, since $\pi < 5\gamma_0$.

(b) *Counting.* Let $c_1, \dots, c_n$ be the distinct inner centres, at distances $r_i \in (\frac{1}{2}, \rho_1]$ from p . Every relevant edge contains the square of its inner centre, so by (a) there are at most $\sum_i D(r_i)$ relevant edges. Put $m_i := \max(2, D(r_i))$, and $R_2 := \rho_1$, $R_m := \min(\rho_1, 1/(2 \sin \frac{(m-1)\gamma_0}{2}))$ for

$m = 3, 4, 5$; then $r_i \leq R_{m_i}$. For $0 < r_1, r_2 < 1$ we have $\partial f / \partial r_1 = (r_1^2 - r_2^2 + 1) / (2r_1^2 r_2) > 0$, and symmetrically, so two inner centres c_i, c_j (at distance ≥ 1) are seen from p at an angle at least $\gamma(m_i, m_j)$, where $\gamma(a, b) := \arccos f(R_a, R_b)$. Order the inner centres by their direction from p . If $n \geq 3$, the angles between consecutive directions add up to 2π , so $\sum_i \gamma(m_i, m_{i+1}) \leq 2\pi$ (indices modulo n); if $n \leq 2$, then $\sum_i m_i \leq 10$. Since R_e does not depend on d , every edge that is relevant for d is relevant for 10^{-4} , so we may take $d = 10^{-4}$, where $\gamma_0 > 41.406^\circ$, $\rho_1 < 0.86611$, $R_3 < 0.75598$, $R_4 < 0.56571$, $R_5 < 0.50396$, and $\gamma(a, b) \geq \gamma(2, 2) > 70.52^\circ$, so $n \leq 5$. A finite check of all sequences $(m_1, \dots, m_n) \in \{2, 3, 4, 5\}^n$, $n \leq 5$, shows that every sequence with $\sum_i m_i \geq 15$ violates these angle conditions, with a margin of more than 25° , and that the only sequences with $\sum_i m_i = 14$ that satisfy them are, up to rotation, $(5, 2, 5, 2)$ (which uses about 359°). The check, in 50-digit arithmetic, is part of the accompanying script. So there are at most 14 relevant edges, and at most 13 unless two inner centres have class 5.

(c) *Two centres of class 5.* For an inner centre c write $\deg(c)$ for the number of relevant edges containing its square; the number of relevant edges is at most $\sum_i \deg(c_i)$, and an edge both of whose members are squares with centres among $c_1, \dots, c_n$ is counted twice in this sum. Suppose that two inner centres, say c_1 and c_3 , have class 5, i.e. $r_1, r_3 \leq R_5 < 0.50396$. Then $1 \leq |c_1 - c_3| \leq 2R_5 < 1.008$ and the angle at p between $c_1 - p$ and $c_3 - p$ is at least $\gamma(5, 5) > 165.6^\circ$, so the angles of the triangle pc_1c_3 at c_1 and at c_3 are acute, and by the law of sines the angle at c_1 between $p - c_1$ and $c_3 - c_1$ is at most $\psi := \arcsin(R_5 \sin(\pi - \gamma(5, 5))) < 7.19^\circ$. If c_3 is the other end point of a relevant edge at c_1 , this edge is counted in $\deg(c_1)$ and in $\deg(c_3)$, so there are at most $\sum_i D(r_i) - 1$ relevant edges. Otherwise let c_j be the other end point of a relevant edge at c_1 . If c_j is a centre, it is the centre of a square other than that of c_3 , so $|c_j - c_3| \geq 1$; as $|c_j - c_1|$ and $|c_3 - c_1|$ lie in $[1, \Lambda)$ (note $2R_5 < \Lambda$), the argument of (a) shows that the directions of $c_j - c_1$ and $c_3 - c_1$ differ by at least γ_0 . If c_j lies on a side, they differ by more than $78^\circ > \gamma_0$, as in (a). Hence the directions of the $c_j - c_1$ lie in the arc of half-width $\arcsin \frac{1}{2r_1} < 90^\circ$ about the direction of $p - c_1$ from which the open arc of half-width γ_0 about the direction of $c_3 - c_1$ has been removed, that is, in two closed arcs, each of length less than $90^\circ + 7.19^\circ - 41.406^\circ < 2\gamma_0$. Each of them contains at most two of these directions, so $\deg(c_1) \leq 4 = D(r_1) - 1$, and again there are at most $\sum_i D(r_i) - 1$ relevant edges. In all cases, by (b), there are at most $\sum_i D(r_i) - 1 \leq \sum_i m_i - 1 \leq 13$ relevant edges if two inner centres have class 5, and at most 13 otherwise. $\square$

The proof of Lemma 4.9 uses no property of G except that the only sides involved are two opposite ones; the bound holds for any family of such pairs at distance $< d$. A computer-assisted refinement of the same counting gives the bound 9, which is sharp for the number of all pairs at distance $< d$ (see the remark after the proof), though not known to be sharp for the edges of a single graph G .

Lemma 4.10 (multiplicity at uncovered points, computer-assisted). *Under the assumptions of Lemma 4.9, every point of the waste lies in R_e for at most $K_w^* := 9$ of the edges e considered there.*

Proof. As in the proof of Lemma 4.9 we may take $d = 10^{-4}$ and $p = 0$, and we keep the notation $\Lambda, \rho_1, \gamma_0$. Let I be the set of all centres within distance ρ_1 of p ; they satisfy $\frac{1}{2} < |c| \leq \rho_1$ and are pairwise at distance at least 1, so $n := |I| \leq 5$. By the proof of Lemma 4.9, every relevant edge contains a square with centre in I . At most E_{II} relevant edges join two squares with centres in I , where E_{II} is the number of pairs in I at distance less than Λ . Every other relevant edge joins a square with centre $c_i \in I$ either to a side of $[0, k]^2$ or to a square whose centre x satisfies $|x| > \rho_1$, $1 \leq |x - c_i| < \Lambda$, $p \in R(c_i, x)$ and $|x - c_j| \geq 1$ for all $c_j \in I \setminus \{c_i\}$, where $R(c_i, x)$ denotes the rectangle R of the two squares with these centres; as in part (a) of the proof of Lemma 4.9, the directions of the vectors $x - c_i$ of these centres are pairwise at least γ_0 apart. Let $D_i^*(I)$ be

the largest number of directions, pairwise at least γ_0 apart, in which such an x exists, and let W_I be the number of $c_i \in I$ that can belong to a relevant edge with a side (at most one per centre, since $\sqrt{2} < k - 2d$). Then the number of relevant edges is at most

$$U(I) := E_{II} + W_I + \sum_{c_i \in I} D_i^*(I).$$

A branch-and-bound computation over the parameters of I shows that a computed upper bound of $U(I)$ is at most 9 on every box of a cover of the parameter domain. Without a side, the centres are written in polar coordinates (r_i, ϑ_i) with $\vartheta_1 = 0 \leq \vartheta_2 \leq \dots \leq \vartheta_n$ (after a rotation and relabelling). If a relevant edge contains a side, its rectangle $R(P, S_0)$ stands on a side of $[0, k]^2$ whose line is at distance $h < d + \frac{\sqrt{2}}{2}$ from p , and the rectangles of all relevant edges with a side stand on one and the same side σ : two of them on opposite sides would force $k < 2d + \sqrt{2}$, which is impossible since $k \geq 2$, and if $R(P, \sigma)$ and $R(P', \sigma')$ with adjacent sides σ, σ' both contained p , then $P \neq P'$ (the rectangle of a square and a side is determined by the square) and, with the corner at the origin, the conditions $|p_x - c_x| \leq \frac{1}{2}$, $0 \leq p_x \leq c'_x < d + \frac{\sqrt{2}}{2}$, $c_x \geq \frac{1}{2}$ and the symmetric ones would give $|c - c'|^2 \leq \frac{1}{2}$, which is impossible. We take σ to be the line $y = -h$ and ignore the other sides (this only drops constraints); then every centre satisfies $y \geq -h + \frac{1}{2}$, no normalisation of ϑ_1 is used, and h is a further parameter. For a box of parameters the directions are divided into 1440 arcs of 0.25° and $|x - c_i|$ into 6 intervals; an arc is discarded only if for each interval one of the conditions $|x| > \rho_1$, $p \in R(c_i, x)$, $|x - c_j| \geq 1$ (and $y \geq -h + \frac{1}{2}$) fails for all parameters in the box and all points of the arc and the interval. Then $D_i^*(I)$ is bounded by the largest number of points, pairwise at least γ_0 apart, in the union of the remaining arcs (the largest of the circular greedy counts started at the beginning of each maximal run of arcs, or $\lfloor 2\pi/\gamma_0 \rfloor$ if no arc is discarded; this gives the maximum, since an optimal set can be rotated backwards, keeping all gaps, until one of its points reaches the start of its run, after which the j -th greedy point is never later than the j -th point of the rotated optimal set); E_{II} by the number of pairs that may be at distance less than Λ ; and W_I by the number of centres that may lie within $d + \frac{\sqrt{2}}{2}$ of the side with $|c_{i,x}| \leq \frac{1}{2}$ and $c_{i,y} \geq 0$. A box is accepted if it contains no admissible configuration (for every parameter of the box two centres are at distance less than 1, or a centre is closer than $\frac{1}{2}$ to the side), or, in the case with a side, if no side edge is possible (then it is covered by the case without a side), or if the bound is at most 9; otherwise it is bisected. For every $n = 1, \dots, 5$, with and without a side, the computation terminated with all boxes accepted (30,365 accepted boxes without a side and 48,308 with a side).

The search itself is not relied upon: it used floating-point interval bounds with a safety margin of 10^{-9} , and every accepted box was re-verified in ball (midpoint-radius interval) arithmetic with rigorous error bounds, using Arb [16], which has been part of FLINT [12] since version 3.0, through `python-flint` [17] (version 0.9.0, with FLINT 3.6.0, at 96-bit working precision), boxes being bisected further where the ball enclosures were too coarse (all 78,673 boxes confirmed, with 59,238 further bisections and largest rigorous bound 9). In this re-verification the arcs of directions are closed and cover the whole circle (the floating-point end point of each arc is replaced, where necessary, by the starting point of the next one, and the floating-point value $2\pi < 2\pi$ by the exact 2π), and every interval of an angle ϑ_i ending at 2π is extended to the exact 2π . The final count of points pairwise at least γ_0 apart in a union of arcs is a finite combinatorial computation on the arc end points; it is done in double precision with γ_0 decreased by 10^{-12} , far above the rounding errors involved. A separate program checks that the accepted boxes leave no gap: it rebuilds the bisection tree from the initial boxes, accepting a node only if it coincides with a recorded box (each used exactly once) or contains no configuration with ordered angles, and confirms in exact rational arithmetic that the volumes of these boxes

add up to that of the initial boxes (this checker reuses the bisection rule of the search, so it is not an independent implementation). The initial boxes and the radial range $[1, \Lambda]$ of the outer centres contain the parameter domain up to the arc $(\widetilde{2\pi}, 2\pi]$ just mentioned, since the floating-point values used for ρ_1 , Λ and $d + \frac{\sqrt{2}}{2}$ are upper bounds of the exact values (checked in high-precision arithmetic); in the re-verification the conditions themselves are evaluated with the exact constants, the floating-point values serving only as end points of the initial boxes. The programs, the lists of accepted boxes and the logs are part of the accompanying material, available at <https://github.com/squarepacker/k2-minus-c> and archived on Zenodo; they were run with Python 3.12.10, python-flint 0.9.0, NumPy 2.5.3 and mpmath 1.4.1. The computation has not been reproduced by an independent implementation. $\square$

Remark 4.11. The proof bounds the number of *all* pairs of squares (or of a square and one of two opposite sides) at distance less than d whose rectangles contain p , without using the graph G , and for this number the bound 9 is sharp: the configuration described in Remark 7.5 below (two axis-parallel squares on both sides of a slit through p , each nearly touching two slightly tilted squares on either side) has 9 such pairs at an uncovered point; we checked this in 80-digit arithmetic for $d = 10^{-6}, 10^{-5}, 10^{-4}$ (and beyond the range of the lemma for $10^{-3}, 10^{-2}$), hence, by monotonicity in d , for every $d \in [10^{-6}, 10^{-4}]$ (the coordinates and the check are part of the accompanying material). We do not know whether 9 edges of a single graph G_h or G_v can meet in this way.

Apart from the definition of R_e , Lemma 4.8(b) is not used below; it bounds the multiplicity at all points, not only at uncovered ones.

Lemma 4.12 (slits). *Let A, B be distinct squares, or let one of them be the left or the right side of $[0, k]^2$, and let E_{AB} be the set of $y \in [0, k]$ such that on the line y both A and B are met, A lies to the left of B , and no square is met strictly between them. For $y \in E_{AB}$ let $g(y)$ be the length of the gap between them, and let $\theta := \theta(A, B)$.*

- (a) *All $y \in E_{AB}$ with $g(y) < 1$ lie in one component of E_{AB} .*
- (b) *On every component J of E_{AB} the function g is convex, and $\int_I g \geq \theta(2 - \theta)|I|^2/4$ for every interval $I \subset J$. In particular, for $\eta > 0$ the set $\{y \in J : g(y) < \eta\}$ is an interval.*

The same holds with the roles of the axes exchanged (vertical lines, A below B , the bottom and the top side).

Proof. Let U be the intersection of the vertical extents of A and B (for a side of the container, its vertical extent is $[0, k]$); it is an interval containing E_{AB} . Let $r_A(y)$ be the largest abscissa of A on the line y ($r_A \equiv 0$ for the left side) and $l_B(y)$ the smallest abscissa of B ($l_B \equiv k$ for the right side); by Lemma 4.2 (with $(s, t) = (x, y)$), $G := l_B - r_A$ is convex on U , and $g = G$ on E_{AB} . If a square Z lies strictly between A and B on one line $y \in U$, then it does so on every line meeting A, B and Z : two disjoint compact convex sets are strictly separated by a line, and if that line is not horizontal their order on every horizontal line meeting both is the same (if it is horizontal they never share a line); when one of A, B is a side of the container this is trivial, since every square lies to the right of the left side and to the left of the right side.

(a) Let $y_1 < y_2$ be points of E_{AB} with $g(y_1), g(y_2) < 1$; by convexity $G < 1$ on $[y_1, y_2]$. Suppose that some $y \in (y_1, y_2)$ is not in E_{AB} . Then a square Z is met strictly between A and B on the line y ; it does not meet the lines y_1 and y_2 , since it would lie between A and B there, so its vertical extent $V(Z)$ lies in $(y_1, y_2) \subset U$. The midpoint y_Z of $V(Z)$ lies in the middle band $\mathcal{M}(Z)$, so the chord of Z at height y_Z has length $\sec a(Z) \geq 1$ (Lemma 3.1(b)), and it lies strictly between A and B . Hence $G(y_Z) > 1$, a contradiction. So $[y_1, y_2] \subset E_{AB}$.

(b) On a component J of E_{AB} , $g = G$ is convex. By Lemma 4.2, it is piecewise linear with at most two break points, at each of which its slope increases by at least 2, and off the break points

$|g'| = |\tan \psi_B - \tan \psi_A| \geq |\psi_B - \psi_A| \geq \theta$, where ψ_A, ψ_B are the angles between the vertical and the sides of A and B bounding the gap (for a side of the container the angle is 0, and $\theta(A, S_0)$ is the inclination of the square). As $\theta \leq \frac{\pi}{4} < 1$, Lemma 4.1 applies on every interval $I \subset J$. Sublevel sets of a convex function on an interval are intervals. $\square$

Lemma 4.13 (cost of a large inclination on a line). *Let $0 < \bar{\beta} \leq 1.5 \cdot 10^{-6}$, and let $Y \subset [0, k]$ and $\beta : Y \rightarrow [0, \bar{\beta}]$ be such that for every $y \in Y$: $\omega(y) < \frac{1}{2}$ and the line y meets a square of inclination $\geq \beta(y)$. Then*

$$\int_Y 2\beta(y) dy \leq AW, \quad A := \frac{4\sqrt{K_w^*/Q_*}}{2 - 3 \cdot 10^{-6}} < 10.6067.$$

Proof. For $y \in Y$ let $X_1, \dots, X_m$ be the squares met by the line, in order, and X_0, X_{m+1} the left and right sides of the container. Each gap is part of the line waste, so it is shorter than $\frac{1}{2}$, and the triangle inequality from the left side to a square of inclination $\geq \beta(y)$ and on to the right side gives $\sum_{j=0}^m \theta(X_j, X_{j+1}) \geq 2\beta(y)$. Put $\theta_{\max} := 2\bar{\beta}$ and $\theta' := \min(\theta, \theta_{\max})$; then also $\sum_j \theta'(X_j, X_{j+1}) \geq 2\beta(y)$, since either some term equals $\theta_{\max} \geq 2\beta(y)$, or no term is capped.

For a pair $e = \{P, Q\}$ of members of $\mathcal{V}_h$, with P the left one, let Y_e be the set of $y \in Y$ on which P and Q are consecutive, and let $\mathcal{E}$ be the finite set of pairs with $|Y_e| > 0$; they are edges of G_h . Write $\theta_e := \theta(P, Q)$, $\theta'_e := \min(\theta_e, \theta_{\max})$, $E_e := E_{PQ} \supset Y_e$, and g_e for the gap function of Lemma 4.12. Integrating over Y ,

$$\int_Y 2\beta \leq \sum_{e \in \mathcal{E}} \theta'_e |Y_e|.$$

Put $d' := \theta_{\max}$ and split $Y_e = Y_e^c \cup Y_e^f$, where $Y_e^c := \{y \in Y_e : g_e(y) < d'\}$. Let $s_e^f := \int_{E_e \cap \{g_e \geq d'\}} g_e$. Since $\theta'_e \leq \theta_{\max} = d'$,

$$\theta'_e |Y_e^f| \leq \frac{\theta_{\max}}{d'} \int_{Y_e^f} g_e \leq s_e^f.$$

Now let $|Y_e^c| > 0$; then the members of e are at distance $< d' \leq 3 \cdot 10^{-6}$. As $g_e < \frac{1}{2}$ on Y_e , Lemma 4.12 shows that Y_e lies in one component J_e of E_e , and that Y_e^c lies in the interval $I_e := \{y \in J_e : g_e(y) < d'\}$, with

$$s_e^c := \int_{I_e} g_e \geq \frac{\theta_e(2 - \theta_e)}{4} |I_e|^2 \geq \frac{\theta'_e(2 - \theta_{\max})}{4} |I_e|^2,$$

because $x(2 - x)$ is increasing on $[0, 1]$. Let f_e be the area of the waste in R_e (notation of Lemma 4.8(b), with $d = d'$). We claim that $f_e \geq Q_* \theta'_e(2 - \theta_{\max})/4$. If $\theta_e \leq \theta_1 := 10^{-5}$, this follows from Proposition 4.5. Otherwise Lemma 4.8(a) gives $f_e \geq \theta_e(2 - \theta_e)/(4q_0) \geq \theta_1(2 - \theta_1)/40$, which exceeds $Q_* \theta_{\max}(2 - \theta_{\max})/4 \geq Q_* \theta'_e(2 - \theta_{\max})/4$ as $\theta_{\max} \leq 3 \cdot 10^{-6}$. (For a square X and a side, $R_e = R(X, S_0)$ may stand on another side of the container; this is harmless, as $\theta(X, S_0) = a(X)$.) Hence

$$\theta'_e |Y_e^c| \leq \theta'_e |I_e| = (\theta'_e |I_e|^2 \cdot \theta'_e)^{1/2} \leq \frac{4}{2 - \theta_{\max}} \left(\frac{s_e^c f_e}{Q_*} \right)^{1/2}.$$

The sets $\{(x, y) : y \in E_e, x \text{ in the gap of } e \text{ on the line } y\}$ are disjoint parts of the waste for different e , and I_e and $E_e \cap \{g_e \geq d'\}$ are disjoint, so $S^c + S^f \leq W$ for $S^c := \sum_e s_e^c$ and $S^f := \sum_e s_e^f$. By Lemma 4.10 (with $d = d'$), every point of the waste lies in R_e for at most K_w^* of the pairs with $|Y_e^c| > 0$, so $\sum f_e \leq K_w^* W$, the sum over these pairs. By the Cauchy-Schwarz

inequality,

$$\sum_{e \in \mathcal{E}} \theta'_e |Y_e| \leq \frac{4\sqrt{K_w^*/Q_*}}{2 - \theta_{\max}} \sqrt{S^e W} + S^f \leq A\sqrt{(W - S^f)W} + S^f \leq AW,$$

the last step because $x \mapsto A\sqrt{(W - x)W} + x$ is decreasing on $[0, W]$ when $A \geq 2$. Finally $4\sqrt{9/0.32}/(2 - 3 \cdot 10^{-6}) < 10.6067$. $\square$

Lemma 4.14 (cost of a large inclination in a column). *Let $0 < \delta \leq 10^{-5}$ and $0 < \bar{\beta} \leq 1.5 \cdot 10^{-6}$. Let $X, X' \subset [0, k]$. For $x \in X$ let $\bar{y}_x \in (0, \frac{k}{2} - 1]$ and $\beta(x) \in [0, \bar{\beta}]$ be such that the segment $\sigma_x = \{x\} \times [0, \bar{y}_x]$ has vertical waste $< \delta$ and meets a square of inclination $\geq \beta(x)$; for $x \in X'$ let $\bar{y}'_x, \beta'(x)$ satisfy the same with the segment $\sigma'_x = \{x\} \times [k - \bar{y}'_x, k]$. Then*

$$\int_X \beta(x) dx + \int_{X'} \beta'(x) dx \leq AW.$$

Proof. For $x \in X$ list the squares met by σ_x from the bottom up to the first one of inclination $\geq \beta(x)$, preceded by the bottom side of the container; for $x \in X'$ list those met by σ'_x from the top down, preceded by the top side. All gaps are $< \delta$, so consecutive members are at distance $< \delta$, and the triangle inequality gives $\sum_j \theta(Y_j, Y_{j+1}) \geq \beta(x)$ (resp. $\beta'(x)$), hence the same for $\theta' := \min(\theta, \bar{\beta})$. Two members that are consecutive on σ_x are consecutive on the whole vertical line through x (a square between them would meet σ_x between them), so the pairs used for a set of x of positive measure are edges of G_v whose members are at distance $< \delta$. Both members of a pair used for some $x \in X$ meet $\mathbb{R} \times [0, \frac{k}{2} - 1]$, and both members of a pair used for some $x \in X'$ meet $\mathbb{R} \times [\frac{k}{2} + 1, k]$; as no square meets both strips, no pair is used in both ways. For a pair e let X_e be the set of x in X , or in X' , at which it is used. Since the gaps are $< \delta < 1$, the vertical version of Lemma 4.12 puts X_e in one component J_e of E_e , and $s_e := \int_{J_e} g_e \geq \theta'_e(2 - \bar{\beta})|J_e|^2/4$. As in the proof of Lemma 4.13, with $d = \delta$ and $\theta_{\max}$ replaced by $\bar{\beta}$, we have $f_e \geq Q_*\theta'_e(2 - \bar{\beta})/4$ (by Proposition 4.5 if $\theta_e \leq \theta_1 = 10^{-5}$, and by Lemma 4.8(a) otherwise, since $\theta_1(2 - \theta_1)/40 \geq Q_*\bar{\beta}(2 - \bar{\beta})/4$). Hence $\theta'_e|X_e| \leq \theta'_e|J_e| \leq \frac{4}{2 - \bar{\beta}}(s_e f_e / Q_*)^{1/2}$, $\sum_e s_e \leq W$ and $\sum_e f_e \leq K_w^* W$ (Lemma 4.10 with $d = \delta$), so the left-hand side is at most $\sum_e \theta'_e|X_e| \leq \frac{4\sqrt{K_w^*/Q_*}}{2 - \bar{\beta}} W \leq AW$. $\square$

5. QUANTIZATION IN COLUMNS

Fix $C := 2$ and $\delta := 10^{-5}$, and put $\alpha(s) := 1/(Cs)$ for $s > 0$.

Definition 5.1. Let $0 < s \leq \frac{k}{2} - 3$. A point $x \in [0, k]$ is *s-rigid from below* if the segment $\sigma_x^s := \{x\} \times [0, s + 2]$ has vertical waste $< \delta$ and meets no square of inclination $\geq \alpha(s)$, and *s-rigid from above* if the segment $\tilde{\sigma}_x^s := \{x\} \times [k - s - 2, k]$ has these two properties. We write NR_s and $\widetilde{\text{NR}}_s$ for the sets of x that are not *s-rigid from below*, respectively from above.

If $s < s'$, then $\sigma_x^s \subset \sigma_x^{s'}$ and $\alpha(s) > \alpha(s')$, so $\text{NR}_s \subset \text{NR}_{s'}$. Moreover, for fixed x the set of $s \in (0, \frac{k}{2} - 3]$ with $x \in \text{NR}_s$ is closed: it is the union of the set where the vertical waste of σ_x^s , a continuous function of s , is $\geq \delta$, and of the sets $\{s : s + 2 \geq b_X, Cs\alpha(X) \geq 1\}$ over the finitely many squares X with $a(X) > 0$ that meet the vertical line through x , where b_X is the lower end point of the chord of X on that line. The same holds for $\widetilde{\text{NR}}_s$.

For a square Z and $x \in \mathcal{M}'(Z)$ let $\beta_Z(x)$ be the height of the lower endpoint of the vertical chord of Z at x . A square $X \neq Z$ is *below Z at x* if X meets $\{x\} \times [0, \beta_Z(x))$; its vertical chord at x is then contained in $[0, \beta_Z(x))$, because it is an interval disjoint from the chord of Z .

Lemma 5.2 (quantization). *Let $s \geq 7.9 \cdot 10^5$ with $s \leq \frac{k}{2} - 3$, let Z be a square with $a(Z) < \alpha(s)$ whose lowest point has height $v_Z \leq s$, and let $x \in \mathcal{M}'(Z)$ satisfy*

- (i) $x \in \mathcal{M}'(X)$ for every square X with $a(X) < \alpha(s)$ that is below Z at x , and
- (ii) x is s -rigid from below.

Then every point of $\mathcal{R}(Z)$ is at distance less than $w_0 := 1.08 \cdot 10^{-5}$ from an integer.

Proof. Write $\beta = \beta_Z(x)$ and $\alpha = \alpha(s) \leq 6.33 \cdot 10^{-7}$; by Lemma 3.1(c), $\beta \leq v_Z + \sin a(Z) < s + \alpha$, so $\{x\} \times [0, \beta) \subset \sigma_x^s$. By (ii) every square below Z at x has inclination $< \alpha$, so by (i) and Lemma 3.1(c) its vertical chord at x has length in $[1, \sec \alpha]$. If n is the number of these squares, their chords and the uncovered part fill $[0, \beta)$, whence $n \leq \beta < n \sec \alpha + \delta$ and $n < s + \alpha$. We use $\sec x - 1 \leq 0.50001 x^2$ for $0 \leq x \leq 10^{-3}$ (the function $(\sec x - 1)/x^2$ has a Taylor series with positive coefficients, so it is increasing on $(0, \frac{\pi}{2})$, and its value at 10^{-3} is less than 0.50001). Now $n(\sec \alpha - 1) \leq (s + \alpha) \cdot 0.50001 \alpha^2 = 0.50001(1 + \alpha/s)/(C^2 s) < 1.59 \cdot 10^{-7}$, so $\beta \in [n, n + 1.0159 \cdot 10^{-5})$. By Lemma 3.1(c), and as $\sin a(Z) < 6.33 \cdot 10^{-7}$, $v_Z \in [\beta - \sin a(Z), \beta] \subset (n - 6.33 \cdot 10^{-7}, n + 1.0159 \cdot 10^{-5})$. The lower ramp of Z lies in $[v_Z, v_Z + \sin a(Z)] \subset (n - 6.33 \cdot 10^{-7}, n + 1.0792 \cdot 10^{-5})$. The upper ramp is $[v_Z + \cos a(Z), v_Z + \cos a(Z) + \sin a(Z)]$; here $v_Z + \cos a(Z) \geq \beta - \sin a(Z) + \cos a(Z) \geq n + \cos \alpha - \sin \alpha > n + 1 - 6.4 \cdot 10^{-7}$ and $v_Z + \cos a(Z) + \sin a(Z) \leq \beta + 1 + \sin a(Z) < n + 1 + 1.0792 \cdot 10^{-5}$. $\square$

For a square X with $a(X) > 0$ and horizontal extent $[x_L, x_R]$, the vertical ramp set of X (Lemma 3.1(c)) consists of two intervals of length $\sin a(X)$: the *left end* $[x_L, x_L + \sin a(X)]$ and the *right end* $[x_R - \sin a(X), x_R]$. They are the projections on the x -axis of the *left side* and the *right side* of X , the sides through the leftmost and the rightmost vertex that make the angle $a(X)$ with the vertical. If $a(X) = 0$ we say that the left end of X is the single point x_L (the abscissa of its left side).

Lemma 5.3 (cracks). *Let $0 < \alpha \leq 10^{-3}$ and $0 < \delta \leq 10^{-3}$ (in this lemma α and δ are arbitrary numbers in these ranges), and let $\sigma = \{x\} \times [0, b]$ be a vertical segment with vertical waste $< \delta$ such that every square meeting σ has inclination $< \alpha$. Let X be a square meeting σ such that x lies in the right end $[\xi_X, \xi_X + \sin a(X)]$ of X (so $0 < a(X) < \alpha$), let $p = (x, h)$ be the point where the vertical line through x meets the right side of X , and assume $h \leq b - \delta$. Then either $x - \xi_X < \delta \tan \alpha (1 + 2.01 \tan^2 \alpha)$, or there is a square $X' \neq X$ that meets σ at a point $q \neq p$ of the vertical segment of length δ that starts at p and leaves X there (so $q \notin X$ and $|q - p| < \delta$), and satisfies*

$$\lambda' \leq x \leq \lambda'' + 2.01 \delta \tan \alpha,$$

where $[\lambda', \lambda'']$ is the left end of X' .

Proof. Write $a := a(X)$, let ς be the right side of X , P_1 its end point at the rightmost vertex (abscissa $\xi_X + \sin a$), and P its other end point (abscissa ξ_X); let P_y denote the height of P . Then ς runs from P_1 in the direction $(-\sin a, u \cos a)$ for some $u \in \{1, -1\}$, and for heights y between those of P_1 and P the right end point of the chord of X at height y is $r(y) := x - u(y - h) \tan a$. Suppose that $x - \xi_X \geq \delta \tan \alpha (1 + 2.01 \tan^2 \alpha)$. Then $|P_y - h| = (x - \xi_X) \cot a \geq \delta(1 + 2.01 \tan^2 \alpha)$, so the heights $h + ut$ with $0 \leq t \leq \delta(1 + 2.01 \tan^2 \alpha)$ lie between those of p and P . Moreover the segment $\{x\} \times \{h + ut : 0 < t < \delta\}$ lies in σ : for $u = 1$ because $h \leq b - \delta$, for $u = -1$ because $P_y \geq 0$. As its uncovered part has length $< \delta$, it contains a point $q = (x, y_q)$, $y_q = h + ut$, $0 < t < \delta$, of some square X' . Since $r(y_q) = x - t \tan a < x$, the point q is not in X , so $X' \neq X$, and $a' := a(X') < \alpha$ because X' meets σ .

The disjoint convex sets X and X' both meet the horizontal line through q , so a line separating them is not horizontal, and on every horizontal line meeting both, X lies to the left of X' . Let $l'(y)$ be the left end point of the chord of X' at height y . Then (A) $l'(y) > r(y)$ for every height

y between those of p and P at which X' is met, and (B) $x - t \tan a = r(y_q) < l'(y_q) \leq x$. Let $\hat{q} := (l'(y_q), y_q)$. If $a' = 0$, then $\hat{q}$ lies on the vertical left side of X' , and (B) gives the claim with $\lambda' = \lambda'' = l'(y_q)$. Let $a' > 0$, and let Q' be the leftmost vertex of X' , so that $\lambda' = Q'_x$. The left boundary $\{(l'(y), y)\}$ of X' consists of the left side, which projects onto $[\lambda', \lambda'']$, and of the other side F' of X' at Q' , which runs from Q' in the direction $(\cos a', v \sin a')$ for some $v \in \{1, -1\}$.

Case 1: $\hat{q}$ lies on the left side. Then $l'(y_q) \in [\lambda', \lambda'']$, and by (B) $\lambda' \leq x < \lambda'' + \delta \tan \alpha$.

Case 2: $\hat{q} = Q' + \mu(\cos a', v \sin a') \in F'$ with $0 < \mu \leq 1$, and $v = -u$. For $0 < w \leq \mu$ the point $\hat{q} - w(\cos a', v \sin a')$ of X' lies at height $y_q + uw \sin a'$. If this height lies between those of p and P , then (A) gives $l'(y_q) - w \cos a' > r(y_q + uw \sin a') = x - (t + w \sin a') \tan a$, that is, $w(\cos a' - \sin a' \tan a) < l'(y_q) - x + t \tan a \leq t \tan a$ by (B). Put $w_* := t \tan a / (\cos a' - \sin a' \tan a)$. If $\mu > w_*$, applying this to all $w \in (w_*, \mu]$ shows that $|P_y - y_q| \leq w_* \sin a'$, hence $|P_y - h| \leq t + w_* \sin a' < \delta(1 + 2.01 \tan^2 \alpha)$, contrary to the first paragraph. So $\mu \leq w_*$, and by (B) $x - \lambda' = (x - l'(y_q)) + \mu \cos a' < t \tan a + t \tan a / (1 - \tan a \tan a') < 2.01 \delta \tan \alpha$, while $x \geq l'(y_q) \geq \lambda'$.

Case 3: $\hat{q} = Q' + \mu(\cos a', v \sin a') \in F'$ with $0 < \mu \leq 1$, and $v = u$. Then the left side of X' runs from Q' in the direction $(\sin a', -u \cos a')$, and $Q'_y = y_q - u \mu \sin a'$, while $h = y_q - ut$. If $\mu \sin a' > t$, the height h lies strictly between Q'_y and y_q , so $l'(h) = l'(y_q) - t \cot a'$; by (A) and (B), $l'(h) > r(h) = x \geq l'(y_q)$, which is absurd. So $\mu \sin a' \leq t$; then the height h lies on the left side of X' , at distance $t - \mu \sin a' < \delta$ from Q'_y , so that $l'(h) \leq \lambda' + \delta \tan \alpha$, and (A) gives $x = r(h) < l'(h) \leq \lambda' + \delta \tan \alpha$; also $x \geq l'(y_q) \geq \lambda'$.

In all cases $\lambda' \leq x \leq \lambda'' + 2.01 \delta \tan \alpha$, and $q \in X' \cap \sigma$ is at distance $t < \delta$ from p . $\square$

Lemma 5.4 (room in a column). *Under the assumptions of Lemma 5.2 on s and Z , let $B_Z = B_Z^{(s)}$ be the set of points $x \in \mathcal{M}'(Z)$ that are s -rigid from below but violate (i). Then $|B_Z| < 0.500018$, so that $G_Z = G_Z^{(s)} := \mathcal{M}'(Z) \setminus B_Z$ has measure greater than $g_0 := 0.49998$.*

Proof. Write $\alpha = \alpha(s) \leq 6.33 \cdot 10^{-7}$, $\varrho := \cos \alpha + \sin \alpha$ and $\mathcal{M}'(Z) = [m_0, m_1]$, so that $m_1 - m_0 = \cos a(Z) - \sin a(Z) \leq 1$; note that the left end of Z ends at m_0 (if $a(Z) = 0$ it is the point m_0 ; if $a(Z) > 0$, then in the coordinates of the proof of Lemma 3.1 the horizontal extent of Z is $[x_0 - \sin a(Z), x_0 + \cos a(Z)]$, its left end is $[x_0 - \sin a(Z), x_0]$ and $\mathcal{M}'(Z) = [x_0, x_0 + \cos a(Z) - \sin a(Z)]$, and likewise in the reflected position). Let $\mathcal{X}_L$ be the set of squares X with $a(X) < \alpha$ that are below Z at some point of $\mathcal{M}'(Z)$ and whose left end is contained in (m_0, m_1) , and let $\mathcal{X}_R$ be the set of squares X with $0 < a(X) < \alpha$ that are below Z at some point of $\mathcal{M}'(Z)$ and whose right end is contained in (m_0, m_1) .

Counting. If $X \in \mathcal{X}_L$ and $y \in \mathcal{M}(X)$ (every y in the vertical extent of X if $a(X) = 0$), the chord of X at height y has its left end point in the left end of X , hence in (m_0, m_1) , and length $\geq 1 \geq m_1 - m_0$, so it contains m_1 . Chords of distinct squares at the same height are disjoint, so the middle bands of the squares in $\mathcal{X}_L$ have pairwise disjoint interiors. A square X below Z at some $x \in \mathcal{M}'(Z)$ meets $\{x\} \times [0, \beta_Z(x))$, and $\beta_Z(x) \leq v_Z + \sin a(Z) < s + \alpha$; so its middle band lies in $[0, s + \alpha + \varrho]$ and has length $\cos a(X) - \sin a(X) > \cos \alpha - \sin \alpha$. Hence $|\mathcal{X}_L| \leq n_* := (s + \alpha + \varrho) / (\cos \alpha - \sin \alpha)$, and in the same way (with m_0 in place of m_1) $|\mathcal{X}_R| \leq n_*$.

Covering. Let $x \in B_Z$. As (i) fails, some square X with $a(X) < \alpha$ is below Z at x and $x \notin \mathcal{M}'(X)$; then $a(X) > 0$ and x lies in the left or in the right end of X . If that end is not contained in (m_0, m_1) , it contains m_0 or m_1 , and $x \in [m_0, m_0 + \alpha) \cup (m_1 - \alpha, m_1]$. If it is the left end and it is contained in (m_0, m_1) , then $X \in \mathcal{X}_L$. If it is the right end and it is contained in (m_0, m_1) , then $X \in \mathcal{X}_R$, and we apply Lemma 5.3 to $\sigma = \sigma_x^s$ (here $b = s + 2$ and $h < \beta_Z(x) < s + \alpha \leq b - \delta$; all squares meeting σ_x^s have inclination $< \alpha$, as x is rigid). Either $x \in [\xi_X, \xi_X + 1.01 \delta \tan \alpha]$, or there is a square X' as in the lemma; let q be the point of $X' \cap \sigma$ found there. If $X' = Z$, then $x \leq m_0 + 2.01 \delta \tan \alpha$. Otherwise $q \notin Z$, and since q lies within δ of

p , which is below the chord $[\beta_Z(x), \beta_Z(x) + \sec a(Z)]$ of Z , the point q lies below that chord; so X' is below Z at x , and $a(X') < \alpha$. If the left end of X' is contained in (m_0, m_1) , then $X' \in \mathcal{X}_L$ and $x \in [\lambda', \lambda'' + 2.01 \delta \tan \alpha]$; otherwise $\lambda' \leq m_0$ or $\lambda'' \geq m_1$; in the first case $\lambda'' < m_0 + \alpha$ and $x < m_0 + \alpha + 2.01 \delta \tan \alpha$, and in the second case $x \geq \lambda' > \lambda'' - \alpha \geq m_1 - \alpha$ (if $a(X') = 0$, then $x \geq \lambda' = \lambda'' \geq m_1$). Altogether,

$$B_Z \subset [m_0, m_0 + 1.01\alpha] \cup [m_1 - \alpha, m_1] \cup \bigcup_{X \in \mathcal{X}_L} [\lambda'_X, \lambda''_X + 2.01 \delta \tan \alpha] \cup \bigcup_{X \in \mathcal{X}_R} [\xi_X, \xi_X + 1.01 \delta \tan \alpha].$$

As $\lambda''_X - \lambda'_X = \sin a(X) < \alpha$ and $\tan \alpha < 1.000001 \alpha$,

$$\begin{aligned} |B_Z| &\leq 2.01 \alpha + n_* \alpha (1 + 3.02 \delta \cdot 1.000001) \\ &\leq 2.01 \alpha + \frac{(1 + (\alpha + \varrho)/s)(1 + 3.04 \cdot 10^{-5})}{C(\cos \alpha - \sin \alpha)} < 0.500018, \end{aligned}$$

since $\alpha \leq 6.33 \cdot 10^{-7}$, $s \geq 7.9 \cdot 10^5$, $\delta = 10^{-5}$ and $C = 2$; and $|\mathcal{M}'(Z)| = \cos a(Z) - \sin a(Z) > 1 - 6.4 \cdot 10^{-7}$. $\square$

Columns from the top. The reflection $(x, y) \mapsto (x, k - y)$ maps $[0, k]^2$ onto itself and closed packings to closed packings; since k is an integer, it maps points at distance less than w_0 from an integer to such points. Applying Lemmas 5.2–5.4 to the reflected packing, we obtain the same statements with “lowest point” replaced by “highest point”, $v_Z \leq s$ by $v'_Z \geq k - s$ for the height v'_Z of the highest point of Z , “below” by “above” (with the upper end point of the vertical chord of Z), and “rigid from below” by “rigid from above”; we denote the corresponding set by $\tilde{G}_Z^{(s)}$.

6. PROOF OF THEOREM 1.1

For $2 \leq k < k_2 = 10^{13}$ we have $\log k < 29.934$, so $0.033 \log k < 0.99 < 1 \leq k^2 - N$ by Corollary 3.3. Let now $k \geq k_2$, and put $\varepsilon := 2 \cdot 10^{-3}$, $\omega_0 := 2 \cdot 10^{-4}$, $y_0 := \sqrt{k}/4 \geq 7.9 \cdot 10^5$, $y_1 := \frac{k}{2} - 3$, $w_0 := 1.08 \cdot 10^{-5}$ and $h_0 := 1 - 2w_0$; recall $C = 2$, $\delta = 10^{-5}$ and $\alpha(s) = 1/(Cs)$ from Section 5, and put $C' := 2\sqrt{2}$ and $\alpha'(y) := 1/(C'y)$ (the threshold for lines). For $y \in (0, k)$ let $d(y) := \min(y, k - y)$ be the distance to the nearer of the bottom and the top side, and for a measurable set $Y \subset \{y : d(y) \geq y_0\}$ write $\ell(Y) := \int_Y dy/d(y)$. Let

$$H := \{y \in (0, k) : y_0 \leq d(y) \leq (1 - \varepsilon)y_1, \{y\} \in [w_0, 1 - w_0]\},$$

$H_b := H \cap (0, \frac{k}{2})$ and $H_t := H \cap (\frac{k}{2}, k)$. Since k is an integer, $y \mapsto k - y$ maps H_b onto H_t and preserves d , so $\ell(H) = 2\ell(H_b)$. If $j_0 := \lceil y_0 \rceil \leq m \leq j_1 - 1$ with $j_1 := \lfloor (1 - \varepsilon)y_1 \rfloor$, then $[m + w_0, m + 1 - w_0] \subset H_b$ and $\int_{m+w_0}^{m+1-w_0} dy/y \geq h_0/(m+1) \geq h_0 \int_{m+1}^{m+2} dy/y$, so

$$\ell(H) \geq 2h_0 \log \frac{j_1 + 1}{j_0 + 1} \geq 2h_0 \log \frac{(1 - \varepsilon)(k/2 - 3)}{\sqrt{k}/4 + 2} \geq h_0(\log k + 1.38228),$$

the last step because $2 \log(2(1 - \varepsilon)) > 1.38229$ and, for $k \geq k_2$, $(1 - 6/k)/(1 + 8/\sqrt{k}) > 1 - 2.6 \cdot 10^{-6}$.

For $y \in H$ we have $\alpha'(d(y)) \leq \alpha'(y_0) = \sqrt{2}/\sqrt{k} \leq 4.48 \cdot 10^{-7}$ and

$$\eta(y) := (k - 1)(\sec \alpha'(d(y)) - 1) \leq 0.50001 \frac{k - 1}{8d(y)^2} \leq 1.00002.$$

Every $y \in H$ is of exactly one of the following kinds: (i) $\omega(y) \geq \omega_0$; (ii) $\omega(y) < \omega_0$ and the line y meets a square of inclination $\geq \alpha'(d(y))$; (iii) $\omega(y) < \omega_0$ and every square meeting the line y has inclination $< \alpha'(d(y))$. Let Y_i, Y_{ii}, Y_{iii} be the corresponding sets.

Kinds (i) and (ii). Since $\omega \geq \omega_0$ on Y_i and $d \geq y_0$ on H , $W \geq \omega_0|Y_i| \geq \omega_0 y_0 \ell(Y_i)$. Lemma 4.13, applied to $Y = Y_{ii}$ with $\beta(y) = \alpha'(d(y)) \leq \bar{\beta} := 4.48 \cdot 10^{-7}$, gives $AW \geq \int_{Y_{ii}} 2\alpha'(d(y)) dy = \frac{2}{C'} \ell(Y_{ii}) = \frac{1}{\sqrt{2}} \ell(Y_{ii})$.

Kind (iii): many non-rigid columns. Let $Y_b := Y_{iii} \cap H_b$ and $Y_t := Y_{iii} \cap H_t$. Let $s \in [y_0, y_1]$ and $\mathcal{W}_s := [(1-\varepsilon)s, s]$, and let $\mathcal{Z}(s)$ be the set of squares Z with $\Phi(Z) \cap Y_b \cap \mathcal{W}_s \neq \emptyset$, where $\Phi(Z) = \{y : 0 < c_Z(y) < 1\}$ is as in Lemma 3.4. For $y \in Y_b \cap \mathcal{W}_s$ we have $d(y) = y$, and Lemma 3.4 with $\alpha = \alpha'(y)$ gives $\sum_{Z: y \in \Phi(Z)} c_Z(y) \geq (1 - \omega_0 - \eta(y))_+$ (the sum is ≥ 0 ; near $d(y) = y_0$ the bound $1 - \omega_0 - \eta(y)$ may be negative); every Z in this sum belongs to $\mathcal{Z}(s)$, meets the line $y \leq s$ (so that the height v_Z of its lowest point satisfies $v_Z \leq s$), and has $0 < a(Z) < \alpha'(y) \leq \alpha'((1-\varepsilon)s)$. Integrating over $Y_b \cap \mathcal{W}_s$ and using $\int_{\Phi(Z)} c_Z = \sin a(Z) \cos a(Z) < \alpha'((1-\varepsilon)s)$ (Lemma 3.4), we get

$$|\mathcal{Z}(s)| \geq C'(1-\varepsilon)s \int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - \eta(y))_+ dy.$$

Let $Z \in \mathcal{Z}(s)$. As $C'(1-\varepsilon) > C$, we have $a(Z) < \alpha'((1-\varepsilon)s) < \alpha(s)$, so Lemmas 5.2 and 5.4 apply to s and Z . Every $x \in G_Z^{(s)}$ is either not s -rigid from below, or it is s -rigid from below and satisfies (i); in the latter case Lemma 5.2 would put every ramp point of Z within w_0 of an integer, which is impossible because Z has a ramp point $y \in H$ (as $\Phi(Z) \subset \mathcal{R}(Z)$). Hence $G_Z^{(s)} \subset \mathcal{M}'(Z) \cap \text{NR}_s$, and $|G_Z^{(s)}| > g_0$ by Lemma 5.4. Every $Z \in \mathcal{Z}(s)$ meets a line in $\mathcal{W}_s$ and has vertical extent < 1.0001 , and a vertical line through x meets every Z with $x \in G_Z^{(s)} \subset \mathcal{M}'(Z)$ in a chord of length ≥ 1 ; these chords are disjoint and lie in an interval of length $< \varepsilon s + 2.01$, so at most $\varepsilon s + 2.01$ of the sets $G_Z^{(s)}$ contain x . Hence

$$(1) \quad \mathcal{N}(s) := |\text{NR}_s| \geq \frac{g_0 |\mathcal{Z}(s)|}{\varepsilon s + 2.01} \geq \frac{g_0 C'(1-\varepsilon)s}{\varepsilon s + 2.01} \int_{Y_b \cap \mathcal{W}_s} (1 - \omega_0 - \eta(y))_+ dy.$$

By the reflection $y \mapsto k - y$ and the versions of Lemmas 5.2 and 5.4 for columns from the top, $\tilde{\mathcal{N}}(s) := |\widetilde{\text{NR}}_s|$ satisfies (1) with $Y_b \cap \mathcal{W}_s$ replaced by $Y_t \cap (k - \mathcal{W}_s)$.

Kind (iii): the cost of non-rigid columns. For $x \in [0, k]$ let $T^*(x)$ be the least $s \in [y_0, y_1]$ with $x \in \text{NR}_s$ (it exists by closedness, if $x \in \text{NR}_{y_1}$; otherwise $T^*(x)$ is undefined), and define $\tilde{T}^*(x)$ in the same way with $\widetilde{\text{NR}}_s$. By monotonicity, $\{x : T^*(x) \leq s\} = \text{NR}_s$ for $s \in [y_0, y_1]$; as these sets are finite unions of intervals, T^* is a Borel function on the Borel set where it is defined, and $\{(x, s) : T^*(x) \leq s\}$ is a Borel set; likewise for $\tilde{T}^*$. If $T^*(x) = \tau$, then either (W) σ_x^τ has vertical waste $\geq \delta$, or (I) it has vertical waste $< \delta$ and meets a square of inclination $\geq \alpha(\tau)$; similarly for $\tilde{T}^*$ and $\tilde{\sigma}_x^\tau$. For each x the segments $\sigma_x^{T^*(x)} \subset \{x\} \times [0, \frac{k}{2} - 1]$ and $\tilde{\sigma}_x^{\tilde{T}^*(x)} \subset \{x\} \times [\frac{k}{2} + 1, k]$ are disjoint, so integrating the vertical waste over x shows that the x of type (W) for T^* and those of type (W) for $\tilde{T}^*$ form two sets of total measure $\leq W/\delta$; on them $\alpha(T^*), \alpha(\tilde{T}^*) \leq \alpha(y_0)$. Lemma 4.14, applied to the x of type (I) with $\bar{y}_x = T^*(x) + 2$, $\beta(x) = \alpha(T^*(x))$, $\bar{y}'_x = \tilde{T}^*(x) + 2$, $\beta'(x) = \alpha(\tilde{T}^*(x))$ and $\bar{\beta} = \alpha(y_0) \leq 6.33 \cdot 10^{-7}$, bounds the rest. Since $\alpha(\tau) \geq \int_\tau^{y_1} ds/(Cs^2)$, Tonelli's theorem gives

$$\left(A + \frac{\alpha(y_0)}{\delta}\right)W \geq \int \alpha(T^*(x)) dx + \int \alpha(\tilde{T}^*(x)) dx \geq \int_{y_0}^{y_1} \frac{\mathcal{N}(s) + \tilde{\mathcal{N}}(s)}{Cs^2} ds.$$

For $y \in Y_b \subset [y_0, (1-\varepsilon)y_1]$ the interval $[y, y/(1-\varepsilon)]$ lies in $[y_0, y_1]$, and

$$\int_y^{y/(1-\varepsilon)} \frac{ds}{s(\varepsilon s + 2.01)} \geq \frac{1}{1 + 2.01/(\varepsilon y_0)} \int_y^{y/(1-\varepsilon)} \frac{ds}{\varepsilon s^2} = \frac{1}{(1 + 2.01/(\varepsilon y_0)) y}.$$

By (1) and Tonelli's theorem again, writing $\int_{Y_b \cap \mathcal{W}_s} f = \int_{Y_b} f(y) \mathbf{1}[y \leq s \leq y/(1-\varepsilon)] dy$, and in the same way for $\tilde{\mathcal{N}}$ (where $d(y) = k - y$),

$$\left(A + \frac{\alpha(y_0)}{\delta}\right)W \geq \frac{g_0 C'(1-\varepsilon)}{C(1+2.01/(\varepsilon y_0))} \int_{Y_{iii}} (1-\omega_0-\eta(y))_+ \frac{dy}{d(y)}.$$

Conclusion. Since $\int_H \eta(y) dy/d(y) \leq 2 \int_{y_0}^{\infty} 0.50001 k dy/(8y^3) = 0.50001 k/(8y_0^2) = 1.00002$ and $(1-\omega_0-\eta)_+ \geq 1-\omega_0-\eta$, adding the three bounds gives (note $(1-\omega_0-\eta)_+ \leq 1$ on $Y_i \cup Y_{ii}$)

$$\begin{aligned} (1-\omega_0)\ell(H) - 1.00002 &\leq \int_H (1-\omega_0-\eta)_+ \frac{dy}{d(y)} \leq \ell(Y_i) + \ell(Y_{ii}) + \int_{Y_{iii}} (1-\omega_0-\eta)_+ \frac{dy}{d(y)} \\ &\leq \left(\frac{1}{\omega_0 y_0} + \frac{C'}{2}A + \frac{C(1+2.01/(\varepsilon y_0))}{C'g_0(1-\varepsilon)}\left(A + \frac{\alpha(y_0)}{\delta}\right)\right)W. \end{aligned}$$

With $y_0 \geq 790569$ (so $1/(\omega_0 y_0) \leq 6.33 \cdot 10^{-3}$ and $2.01/(\varepsilon y_0) \leq 1.272 \cdot 10^{-3}$), $A < 10.6067$, $\alpha(y_0)/\delta \leq 0.0633$ and $g_0 = 0.49998$, the bracket is less than

$$6.33 \cdot 10^{-3} + \sqrt{2} \cdot 10.6067 + \frac{1.001272}{\sqrt{2} \cdot 0.49998 \cdot 0.998} (10.6067 + 0.0633) = 30.1462 \dots < 30.147,$$

so, as $(1-\omega_0)h_0 \geq 0.99977$, $W \geq (0.99977(\log k + 1.38228) - 1.00002)/30.147 \geq (0.99977 \log k + 0.3819)/30.147$, which exceeds $0.0331 \log k$. This proves Theorem 1.1 with $\kappa = 0.033$. $\square$

7. REMARKS

Remark 7.1 (where integrality and closedness enter). Closedness is used in Lemma 3.2 (hence in Lemma 3.4 and Corollary 3.3), in Propositions 4.3 and 4.5 and Lemma 4.12 (separation of disjoint squares by a line), in Lemmas 4.8, 4.9 and 4.10 (centres of disjoint squares are at distance at least 1), in Lemma 4.7 (consecutive squares do not touch), and in Lemma 2.1, which ties closed packings of $[0, k]^2$ to $s(n) < k$; for squares with disjoint interiors in a square of side exactly k , the $k \times k$ grid shows that nothing can be said. The integrality of the side enters through Lemma 3.2 (the bound $k-1$ on long chords) and through the columns hanging from the top side: the reflection $y \mapsto k-y$ preserves the heights close to integers only because k is an integer. The quantization of Lemma 5.2 itself only uses one side and the unit size of the squares.

Remark 7.2 (large inclinations). The same line argument without the column analysis gives the following dichotomy. Let $k \geq 3 \cdot 10^{10}$ and $\alpha := 1/(4\sqrt{k}) \leq 1.45 \cdot 10^{-6}$. The lines with $\omega(y) \geq \frac{1}{2}$ have measure at most $2W$; by Lemma 4.13 with $\beta \equiv \alpha$, the lines with $\omega(y) < \frac{1}{2}$ that meet a square of inclination $\geq \alpha$ have measure at most $AW/(2\alpha)$; and on the remaining lines Lemma 3.4 gives short chords of total length $> 1 - \omega(y) - 0.0313$. Integrating,

$$0.9687k \leq (2.94 + 1.94A\sqrt{k})W + \sum_{0 < a(S) < \alpha} \sin a(S) \cos a(S).$$

Thus a closed packing with $W = o(\sqrt{k})$ must contain at least $(3.87 - o(1))k^{3/2}$ squares of inclination in $(0, \alpha)$.

Remark 7.3 (the threshold and the constant as functions of k). Put $F(k) := (0.99977 \log k + 0.3819)/30.147$ for $k \geq k_2$ and $F(k) := 1$ for $2 \leq k < k_2$, so that every closed packing satisfies $W = k^2 - N \geq \max(1, F(k))$, and let $\kappa(k_1) := \inf_{k \geq k_1} \max(1, F(k))/\log k$ be the largest constant that these bounds certify for all $k \geq k_1$. It is non-decreasing in k_1 . The expression $(0.99977 \log k + 0.3819)/30.147$ exceeds 1 already for $\log k > 29.772$, below $\log k_2 = 29.9336 \dots$; so for integers $2 \leq k < k_2$ the certified ratio is $1/\log k \geq 1/\log(k_2 - 1) = 0.033407 \dots$, while for $k \geq k_2$ the ratio $F(k)/\log k$ decreases from $0.03358 \dots$ to its limit $0.99977/30.147 = 0.033163 \dots$

Hence $\kappa(k_1) = 0.033163\dots$ for every $k_1 \geq 2$, the infimum being approached only as $k \rightarrow \infty$; the threshold k_2 does not limit the constant. The limit $1/(2\sqrt{2}A_\infty) = \sqrt{Q_*/K_w^*}/(4\sqrt{2}) \approx 0.033333$, where $A_\infty = 2\sqrt{K_w^*/Q_*}$, is the most that this architecture can give with $K_w^* = 9$ and $Q_* = 0.32$. For comparison, the bound $F(k) = (0.99996 \log k + 0.386)/36.081$ of the previous draft (with $K_w = 13$ and $k_2 = 10^{16}$) gave $\kappa = 0.027$, the bound $F(k) = (0.99996 \log k + 0.386)/37.442$ of an earlier draft (with $K_w = 14$) gave $\kappa = 0.026$, the bound $F(k) = (0.999996 \log k + 0.386)/51.963$ of an earlier draft (with the multiplicity $K = 27$ in place of K_w) exceeded 1 only for $\log k > 51.58$ and gave $\kappa = 0.019$, and the bound $F(k) = (0.9987(\log k - 1.3884) - 0.0313)/84.05$ of an earlier draft exceeded 1 only for $\log k > 85.58$, and the constant it gave for all $k \geq 2$ was $1/85.58 \approx 0.0117$.

Remark 7.4 (the rate). The proof gives $c^*(k) \gg \log k$, while the upper bounds give $c^*(k) = O(k^{3/5})$. Several heuristics (balancing the cost of inclinations against the quantization near the walls, and numerical linear-programming bounds for periodic covering measures) suggest that the truth may be of order $\sqrt{k}$; we leave this open. The constants in Theorem 1.1 are far from useful for concrete values: since W is an integer, the second bound of the theorem yields $s(k^2 - c) = k$ for every $c \geq 2$ as soon as $\log k > 30.16c - 0.38$ (a threshold above $\log k_2$, so that the second bound applies); for instance (with the exact threshold $(30.147c - 0.3819)/0.99977$) it yields $s(k^2 - 4) = k$ for $\log k > 120.24$ (compare the computer-assisted, unrefereed announcement of $s(k^2 - 4) = k$ for all $k \geq 5$ in [10]), and it says nothing about $s(12)$ or about any k within computational reach.

Remark 7.5 (the constant). An earlier draft of this argument, which used the lemma of [23] with its stated constant 10^{-10} and coarser bookkeeping, obtained only $\kappa \approx 6 \cdot 10^{-17}$, and previous drafts of this paper obtained $\kappa = 3 \cdot 10^{-4}$, $1.9 \cdot 10^{-3}$, 0.011 , 0.019 , 0.026 and 0.027 . The present constant comes from Proposition 4.3 (convex gaps, and cuts only where the set of squares met by the chords changes) and its refinement Proposition 4.5 for nearly parallel pairs, which replaces the factor $1/q_0 = 1/10$ by $Q_* = 0.32$; from the angular argument of Lemma 4.9 and its computer-assisted refinement Lemma 4.10, which bound the multiplicity of the rectangles at uncovered points, the only ones that matter, by $K_w = 13$ and $K_w^* = 9$, the latter sharp for all pairs at distance $< d$ (the planarity argument of Lemma 4.8(b), with Fodor's theorem [13] and Oler's inequality [22, 14], gives $K = 27$ at every point and is no longer needed); from Lemma 4.12(a), by which each pair is charged to a single slit; from capping the inclinations and splitting off the lines with a wide gap in Lemma 4.13; from Lemma 3.4, by which the short chords on a line with little waste add up to almost a full unit (this halves the cost of the columns); from Lemma 5.3, by which a rigid column through the end of a square passes through a tight crack, so that the exceptional columns under a square have measure about $1/C$ rather than $2/C$; from measuring heights from both the bottom and the top side, which doubles the logarithmic range of scales; and from working with a continuum of scales instead of dyadic bands. With $g_0 \approx 1 - \frac{1}{C}$ the bracket in the proof is about $(\frac{C'}{2} + \frac{C}{C'(1-1/C)})A$, which is minimal, equal to $2\sqrt{2}A$, at $C = 2$, $C' = 2\sqrt{2}$. The factor $\sqrt{K_w^*/Q_*} \approx 5.3$ in A comes from using Propositions 4.3 and 4.5 for pairs that are consecutive only on a short set of lines, typically at corners where several squares meet, whereas in simple configurations, such as a single slightly tilted square in a grid, the ratio of $\int_Y 2\beta$ to W in Lemma 4.13 is of order 1. A lemma charging such short-lived pairs to the slits of neighbouring long-lived pairs with a bounded constant would improve κ further; we have not been able to prove one. In numerical experiments (randomized dense packings and targeted searches) the largest multiplicity of the rectangles R_e found by the accompanying search programs was 5 (against 27; an earlier search, whose program was not preserved, reported 8); and the largest number of pairs of squares at distance $< 10^{-4}$ (whether

or not they are edges of G_h or G_v) whose rectangles $R(P, Q)$ contain a common uncovered point was 9, equal to K_w^* , in a degenerate configuration: two axis-parallel squares on both sides of a slit of width about 10^{-20} through the point, each nearly touching two slightly tilted squares on either side (random searches found at most 6). We also found no configuration with $\sum \lambda_r^2 < \frac{1}{3}$ in Proposition 4.5 (against 0.32; the smallest value found was 0.333335). The search programs and their results are part of the accompanying material. With $Q_* = \frac{1}{3}$ the same proof would give, asymptotically, $\kappa \approx 0.0338$. Only the numerical constants depend on the computer-assisted Lemma 4.10: with the analytic bound $K_w = 13$ of Lemma 4.9 in place of K_w^* , the proof of Section 6 gives $A < 12.7476$, a bracket less than 36.212, and $\kappa = 0.027$. Even with A of order 1, this architecture would limit κ to roughly $\frac{1}{2\sqrt{2}A}$, so that a bound useful for concrete k would require a different rate rather than better constants.

Use of AI. Claude (Anthropic) was used extensively in developing the arguments, writing the text and the programs (including the computation for Lemma 4.10), and in checking them. It is not an author; the author takes full responsibility for the contents. The paper has not been refereed.

Licence. This text is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0); the accompanying programs and data are distributed under the MIT licence.

REFERENCES

- [1] M. Z. Arslanov, H. D. Bui, Note on “Efficient packings of unit squares in a large square”, *Discrete Comput. Geom.* **76** (2026), no. 2, 805–812, doi:10.1007/s00454-025-00767-w.
- [2] M. Z. Arslanov, S. A. Mustafin, Z. K. Shangitbayev, Improved packings of $n(n-1)$ unit squares in a square, *Electron. J. Combin.* **28**(4) (2021), #P4.22, doi:10.37236/8586.
- [3] W. Bentz, Optimal packings of 13 and 46 unit squares in a square, *Electron. J. Combin.* **17** (2010), #R126, doi:10.37236/398.
- [4] W. Bentz, Optimal packings of 22 and 33 unit squares in a square, arXiv:1606.03746 (2016).
- [5] H. D. Bui, Square packing with asymptotically smallest waste only needs good squares, arXiv:2504.09489 (2025).
- [6] H. D. Bui, Square packing with $O(x^{0.6})$ wasted area, arXiv:2508.04603v2 (2025, revised 2026).
- [7] chelokot, *square-packing-archive*, GitHub repository, <https://github.com/chelokot/square-packing-archive>, files docs/nagamochi-score-counterexample.md and docs/nagamochi-compensation-proof.md, commit 753079e (accessed 3 October 2026).
- [8] F. Chung, R. Graham, Packing equal squares into a large square, *J. Combin. Theory Ser. A* **116** (2009), no. 6, 1167–1175, doi:10.1016/j.jcta.2009.02.005.
- [9] F. Chung, R. Graham, Efficient packings of unit squares in a large square, *Discrete Comput. Geom.* **64** (2020), no. 3, 690–699, doi:10.1007/s00454-019-00088-9.
- [10] E. Daniel, *square-packing*, GitHub repository, <https://github.com/evand/square-packing>, directories s12/certificates/k2m3 and s12/certificates/k2m4 and file s12/search/FRIEDMAN.md, commit 7ff3b21 (accessed 5 October 2026).
- [11] P. Erdős, R. L. Graham, On packing squares with equal squares, *J. Combin. Theory Ser. A* **19** (1975), no. 1, 119–123, doi:10.1016/0097-3165(75)90099-0.
- [12] The FLINT team, *FLINT: Fast Library for Number Theory*, version 3.6.0 (2026), <https://flintlib.org>.
- [13] F. Fodor, The densest packing of 12 congruent circles in a circle, *Beiträge Algebra Geom.* **41** (2000), no. 2, 401–409.
- [14] J. H. Folkman, R. L. Graham, A packing inequality for compact convex subsets of the plane, *Canad. Math. Bull.* **12** (1969), no. 6, 745–752, doi:10.4153/CMB-1969-096-7.
- [15] E. Friedman, Packing unit squares in squares: a survey and new results, *Electron. J. Combin.*, Dynamic Survey DS7 (1998, revised 2009), doi:10.37236/28.
- [16] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017), no. 8, 1281–1292, doi:10.1109/TC.2017.2690633.

- [17] F. Johansson, O. Benjamin, et al., *python-flint*, version 0.9.0 (2026), <https://github.com/flintlib/python-flint>.
- [18] H. Karakuş, A counterexample to Nagamochi's scoring lemma and a new rectangle packing bound, arXiv:2609.37410 (2026).
- [19] M. J. Kearney, P. Shiu, Efficient packing of unit squares in a square, *Electron. J. Combin.* **9** (2002), #R14, doi:10.37236/1631.
- [20] R. McClenagan, Optimally packing a large square by unit squares, arXiv:2602.01484 (2026).
- [21] H. Nagamochi, Packing unit squares in a rectangle, *Electron. J. Combin.* **12** (2005), #R37, doi:10.37236/1934.
- [22] N. Oler, An inequality in the geometry of numbers, *Acta Math.* **105** (1961), 19–48, doi:10.1007/BF02559533.
- [23] K. F. Roth, R. C. Vaughan, Inefficiency in packing squares with unit squares, *J. Combin. Theory Ser. A* **24** (1978), no. 2, 170–186, doi:10.1016/0097-3165(78)90005-5.